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# Bayesian model averaging based reliability analysis method for monotonic degradation dataset based on inverse Gaussian process and Gamma process

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ABSTRACT

A Bayesian model averaging based reliability analysis method for monotonic degradation modeling and inference is proposed in this paper. Considering the model uncertainty, the Bayesian model averaging method is applied to combine the candidate monotonic processes, specifically the Gamma process and inverse Gaussian process. To evaluate the population reliability, the unit-to-unit variations and heterogeneities within product population are highlighted, so the random effects of both the model parameters and model probabilities are taken in to account. The fully Bayesian inference is applied to estimate distribution hyper-parameters, in which the priors are obtained by moment estimation combined with maximum-likelihood estimation. The proposed Bayesian model averaging based reliability analysis method is verified using previously published GaAs laser degradation dataset. The results indicate that the proposed Bayesian model averaging based method provides flexibility when evaluating the population reliability.

## 1. Introduction

Manufactured products, such as LED-based lights [1], lithium-ion batteries [2], aircraft equipments [3], etc., have seen a growing demand for increased life and improved reliability. Reliability tests are commonly performed before the products are placed into production in order to evaluate their life and reliability. Life data and degradation data are two types of information that can be obtained from the reliability tests [4]. Compared with the life data, the degradation data is easier to obtain, so the reliability evaluation methods based on degradation modeling have received wider attention [5].

Many degradation estimation methods have been proposed to estimate reliability and lifetime based on degradation data [6,7]. One of the most frequently applied methods is degradation-path-based modeling method, which has been used in system reliability modeling [8], degradation analysis [9], and in degradation test planning [10]. However, the degradation-path-based modeling method cannot include the time-dependent structures. [11].

Another approach is the application of the stochastic-process-based modeling method, where the Wiener process, Gamma process, and inverse Gaussian (IG) process are the three most commonly used types of stochastic processes. Typically, the Wiener process is used to analyze the non-monotonic degradation processes, while the Gamma process and IG process are suitable for analyzing monotonic degradation processes [12–14]. It has been demonstrated that the IG process is applicable to a number of different applications such as energy pipelines [15], GaAs lasers [16], etc.

Typically, more than one stochastic process can be used to fit degradation data set. A number of methods have been proposed to handle these problems, which are commonly called model uncertainty [17]. In addition to the Bayesian information criterion (BIC), the Akaike information criterion (AIC) for selection of the most suitable model [18,19] was developed by Sugiura [20] and implemented in practice by Hurvich and Tsai [21].

However, all of the current information criteria can only determine the best model and cannot be applied to analyze the model uncertainty. Hence, several methods have been proposed to analyze the model uncertainty, where three of the most commonly used methods are Dempster–Shafer (D-S) evidence theory, adjustment factor method, and model averaging method.

The D-S evidence theory is based on the plausibility and belief functions which are introduced to account for the epistemic uncertainty [22,23]. The method is commonly used in reliability engineering to handle model uncertainty [24–26]. However, it is difficult to elicit the belief values based on the expert knowledge in order to evaluate the model possibilities, especially in stochastic-process-based degradation modeling.

The adjustment factor method and model averaging methods have

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**Nomenclature**

*A. Acronyms*

| | |
|---|---|
| AIC | Akaike information criterion. |
| BMA | Bayesian model averaging. |
| BIC | Bayesian information criterion. |
| CDF | Cumulative density function. |
| D–S | Dempster–Shafer. |
| IG | Inverse Gaussian. |
| MCMC | Markov chain Monte Carlo. |
| ME | Moment estimation. |
| MLE | Maximum-likelihood estimation. |
| $M_{\text{REPMP}}$ | The BMA based model considering the random effects of parameters and model probabilities. |
| $M_{\text{REP}}$ | The BMA based model considering the random effects of parameters. |
| $M_{\text{REMP}}$ | The BMA based model considering the random effects of model probabilities. |
| $M_{\text{NRE}}$ | The BMA based model without considering the random effects. |
| $M_1$ | IG process. |
| $M_2$ | Gamma process. |
| PDF | Probability density function. |
| TN | Truncated normal. |
| Sd | Standard deviation. |
| *s*-independent | Statistically independent. |
| *s*-credibility | Statistically credibility. |

*B. Notation*

| | |
|---|---|
| $\Lambda(t)$ | Monotonic increasing function. |
| $\Phi(\cdot)$ | CDF of standard Gaussian distribution. |
| $L(\cdot)$ | Likelihood function. |
| $l(\cdot)$ | Log-likelihood function. |
| $\phi(\cdot)$ | PDF of standard Gaussian distribution |
| $Y(t)$ | Product degradation path. |
| $\mathbf{Y}_{mn}$ | Degradation observation dataset, which has $m$ tested samples and $n$ observations for each sample. |
| $y_{i,j}$ | Degradation observation at $j$th measurement for $i$th tested sample. |
| $\boldsymbol{Y}_i$ | Vector of degradation observations for $i$th tested sample. |
| $D$ | Predefined failure threshold. |
| $\Gamma(\cdot)$ | Gamma function. |
| $\text{TN}(\cdot,\cdot,\cdot,\cdot)$ | Truncated normal distribution. |
| $\varsigma(\cdot,\ \cdot)$ | Lower incomplete Gamma function |
| $\Gamma(\cdot,\ \cdot)$ | Upper incomplete Gamma function. |
| $\theta_{\text{IG}}, \theta_{\text{Ga}}, \theta_{\text{SP}}, \theta_M, \boldsymbol{\theta}$ | Vector of distribution parameters. |
| $\theta^{\text{H}}_{\text{IG}}, \theta^{\text{H}}_{\text{Ga}}, \theta^{\text{H}}_{\text{SP}}, \theta^{\text{H}}_{\text{M}}, \theta^{\text{H}}$ | Vector of distribution hyper-parameters. |
| $T$ | Observation time interval. |
| $\hat{\theta}_{\text{IG},i}, \hat{\theta}_{\text{Ga},i}$ | Vector of parameters' MLE results for $i$th tasted sample. |
| $E(\cdot)$ | Expectation. |
| $Var(\cdot)$ | Variance. |
| $\hat{\theta}^{\text{H}}_{\text{IG}}, \hat{\theta}^{\text{H}}_{\text{Ga}}$ | Vector of ME results of hyper-parameters. |
| $\text{Uniform}(\cdot,\ \cdot)$ | Uniform distribution. |
| $\text{IG}(\cdot)$ | IG distribution. |
| $\text{Ga}(\cdot)$ | Gamma distribution. |
| $f(\cdot), f_{\text{IG}}(\cdot), f_{\text{Ga}}(\cdot)$ | PDF. |
| $R(\cdot)$ | Reliability function. |
| $\pi(\cdot)$ | Prior. |
| $\Gamma'(x)$ | The derivative of Gamma function. $k$ |
| $p_{\text{M}_1}, p_{\text{M}_2}$ | The number of parameters in the model. Model probability for IG process and Gamma process. |
| $a_{\text{IG}}, b_{\text{IG}}, \mu_{\text{IG}}, \lambda_{\text{IG}}$ | The parameters for IG distribution. |
| $a_{\text{Ga}}, b_{\text{Ga}}, , \lambda_{\text{Ga}}$ | The parameters for Gamma distribution. |
| $\delta_{\mu_{\text{IG}}}, \gamma_{\mu_{\text{IG}}}, \delta_{\lambda_{\text{IG}}}, \gamma_{\lambda_{\text{IG}}}$ | The hyper-parameters for IG process. |
| $\delta_{\mu_{\text{Ga}}}, \gamma_{\mu_{\text{Ga}}}, \delta_{\lambda_{\text{Ga}}}, \gamma_{\lambda_{\text{Ga}}}$ | The hyper-parameters for Gamma process. |
| $P(\cdot)$ | Probability. |
| $N$ | The number of the samples in the Monte Carlo simulation. |
| $T_{\text{up}}$ | The simulation length of degradation time. |
| $\Delta t$ | The simulation step size of degradation time. |

been studied and mutually compared in Ref. [27]. The model uncertainty in the case of adjustment factor method is resolved by identifying the reference model, and then the predictions from the reference model are modified by the adjustment factor of a given distribution. A number of studies were performed to make this method more applicable to reliability evaluation. For example, in order to consider difference between the experimental and field conditions, the calibration factors are introduced to modify the predictions from the reference model [28,29], where the two sets of data are fused based on Bayesian evaluation method. Additional studies focus on adapting the adjustment factor method to handle the model uncertainty, [30–32].

The model averaging method is widely used in engineering analysis [33–35] and reliability engineering [36–39]. In reliability engineering, the model averaging method has been fully extended into the Bayesian model averaging (BMA) method by combining the model averaging method with the Bayesian inference method [36–39]. In order to access the reliability evaluation with multi model fusion, based on Bayesian framework, this method presents the information using random variables and fuses the parameter information from each corresponding model, by assuming the adaptive models are a form of reliability evaluation information.

Hoeting et al. [40] applied the BMA method in degradation dataset analysis and compared the BMA based method with the model selection based method. The authors concluded that the model uncertainty and the over-confident inference problem can be avoided by using the BMA method. In Ref. [41], both the parameter uncertainty and model uncertainty are considered. The study separates available information into the information about adaptive models and the information from adaptive models and applies general Bayesian framework to treat candidate models as a form of information sources. The authors concluded that both the model uncertainty and parameter uncertainty are need to be considered in fire risk analysis. In Ref. [42], the BMA method is combined with the survival modelling approach to predict the water main failures. Based on the BMA method, the prediction model of water main failures was developed to improve the understanding of the failure process. The results show that the BMA based modeling method is better than cox-proportional hazard model since it considers the model uncertainty. Liu et al. [43] applied the BMA method to address the stochastic degradation model uncertainty in constant stress accelerated degradation dataset analysis. The three most widely used stochastic processes, the IG process, Gamma process, and Wiener process, were selected as the candidate models. The simulation studies were carried out to compare the BMA based method with the model selection based method, where the focus was on the ability of *s*-credibility degradation data analysis. According to the simulation results, the stochastic degradation model uncertainty are successfully captured by the proposed BMA based method. However, in all of above discussed papers, the random effects caused by the unit-to-unit variations and heterogeneities within product population were neglected, especially the random effects of the model possibilities.

In order to verify applicability of Bayesian method in engineering practice Peng et al. [44–46] performed series of studies on Bayesian degradation analysis. The population reliability was evaluated based on the unit-to-unit variations and heterogeneities within product population. The authors argued that the heterogeneities are handled well by introducing the random effects for population reliability evaluation. Several actual examples were presented to validate the effectiveness of the method such as heavy-duty machine tool's spindle system and oil debris. In addition to these studies, the necessity of considering the unit-to-unit variations and heterogeneities in population reliability evaluation is also evaluated [47–49]. Since there are no BMA based reliability analysis methods which consider the unit-to-unit variations and heterogeneities within product population, an improved BMA based analysis method for population reliability evaluation is proposed in this paper.

A BMA based reliability analysis method for monotonic degradation datasets based on IG process and Gamma process with random effects is proposed in this paper. The main distinguishing features of the proposed method compared to the previously published studies are as follows. In order to consider the model uncertainty, the BMA method is applied to combine the candidate processes, Gamma process and IG process. Considering the unit-to-unit variations and heterogeneities within product population, the distribution hyper-parameters are introduced in the degradation processes and model possibilities and estimated based on Bayesian inference method.

This paper is organized as follows. In Section 2, the monotonic degradation process is modeled separately by the IG process model with random effects and Gamma process model with random effects. In Section 3, the integrated BMA based model method is introduced to analyze the model uncertainty. In order to capture the effects of unit-to-unit variations and heterogeneities within product population, the random effects are accounted in the model probabilities by truncated normal (TN) distribution with hyper-parameters. In Section 4, the Bayesian inference is used to estimate the distribution hyper-parameters. The priors are obtained from the degradation dataset by ME combined with MLE. Section 5 demonstrates effectiveness of the proposed method and compares it with other methods. Finally, discussion and conclusions are presented in Section 6.

## 2. System description

In this section, the unit-to-unit variations and heterogeneities within product population are considered, the IG process and Gamma process, both with random effects, are described, and examples of the linear degradation processes are presented, as shown in Fig. 1, where $Y(t)$ is the degradation path, $\mathrm{Ga}(\cdot)$ means Gamma distribution, $\mathrm{IG}(\cdot)$ means IG distribution, $\theta_{\mathrm{IG}}$ and $\theta_{\mathrm{Ga}}$ are vectors of distribution parameters for IG process and Gamma process respectively, $\theta_{\mathrm{SP}}$ is the vector of distribution parameters for stochastic processes, $\boldsymbol{\theta}$ is the vector of distribution parameters, $p_{\mathrm{M_1}}$ is the model probability for IG process, $\theta_{\mathrm{M}}^{\mathrm{H}}$ is the vector of hyper parameter for model probability, $\theta_{\mathrm{SP}}^{\mathrm{H}}$ is the vector of model hyper parameters, and $\theta^{\mathrm{H}}$ is the vector of hyper parameters.

### *2.1. IG process with random effects*

The probability density function (PDF) of IG distribution $y \sim \mathrm{IG}(a_{\mathrm{IG}}, b_{\mathrm{IG}})$, $a_{\mathrm{IG}}, b_{\mathrm{IG}} > 0$, where mean and variance are $a_{\mathrm{IG}}$ and $a_{\mathrm{IG}}^3/b_{\mathrm{IG}}$, is defined as

$$f_{\mathrm{IG}}(y|a_{\mathrm{IG}}, b_{\mathrm{IG}}) = \sqrt{\frac{b_{\mathrm{IG}}}{2\pi y^3}} \exp\left[-\frac{b(y - a_{\mathrm{IG}})^2}{2{a_{\mathrm{IG}}}^2 y}\right], y > 0 \tag{1}$$

Its cumulative density function (CDF) is defined as

$$F_{\mathrm{IG}}(y|a_{\mathrm{IG}}, b_{\mathrm{IG}}) = \Phi\left[\sqrt{\frac{b_{\mathrm{IG}}}{y}}\left(\frac{y}{a_{\mathrm{IG}}} - 1\right)\right] + \exp\left(\frac{2b_{\mathrm{IG}}}{a_{\mathrm{IG}}}\right)\Phi\left[-\sqrt{\frac{b_{\mathrm{IG}}}{y}}\left(\frac{y}{a_{\mathrm{IG}}} + 1\right)\right] \tag{2}$$

where $\Phi(\cdot)$ is the CDP of standard Gaussian distribution.

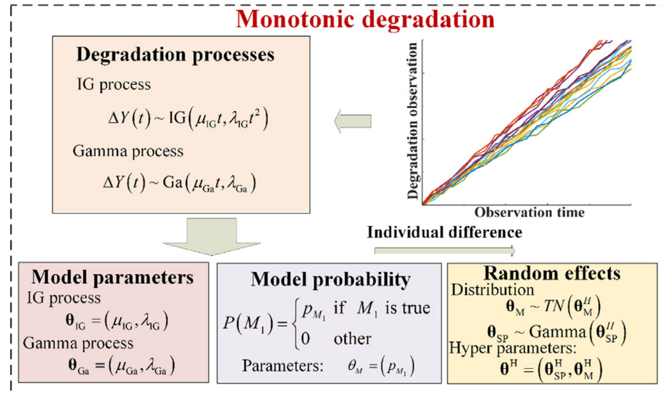


**Fig. 1.** Schematic description of the presented system.

A degradation process which can be described as IG process, has the following properties.

(a) The initial degradation $Y(0) = 0$ with probability 100%.
(b) $Y(t)$ has independent increments, e.g. $Y(t_4)$ - $Y(t_3)$ and $Y(t_2)$ - $Y(t_1)$ are independent only if $t_4 > t_3 > t_2 > t_1$.
(c) The increments $\Delta Y(t) = Y(t + \Delta t) - Y(t)$ follow IG distribution, as

$$\Delta Y(t) \sim \mathrm{IG}(\mu_{\mathrm{IG}}\Delta\Lambda(t), \lambda_{\mathrm{IG}}\Delta\Lambda^2(t)) \tag{3}$$

with parameters $\theta_{\mathrm{IG}} = (\mu_{\mathrm{IG}}, \lambda_{\mathrm{IG}})$, where $\mu_{\mathrm{IG}} > 0$, $\lambda_{\mathrm{IG}} > 0$, and $\Delta\Lambda(t) = \Delta\Lambda(t + \Delta t) - \Lambda(t)$, and where$\Lambda(t)$ is a monotonic increasing function.

The linear degradation processes are considered in this work, which allows the increments to be written as follows

$$\Delta Y(t) \sim \mathrm{IG}(\mu_{\mathrm{IG}}t, \lambda_{\mathrm{IG}}t^2) \tag{4}$$

Hence, the product degradation path $Y(t)$ has the following PDF,

$$f_{\mathrm{IG}}(y(t)|\mu_{\mathrm{IG}}t, \lambda_{\mathrm{IG}}t^2) = \sqrt{\frac{\lambda_{\mathrm{IG}}t^2}{2\pi y^3(t)}} \exp\left[-\frac{\lambda_{\mathrm{IG}}(y(t) - \mu_{\mathrm{IG}}t)^2}{2\mu_{\mathrm{IG}}^2 y(t)}\right] \tag{5}$$

Given a predefined failure threshold $D$, the reliability can be given as

$$R_{\mathrm{IG}}(t|\mu_{\mathrm{IG}}t, \lambda_{\mathrm{IG}}t^2) = \Phi\left[\sqrt{\frac{\lambda_{\mathrm{IG}}}{D}}\left(\frac{D}{\mu_{\mathrm{IG}}} - t\right)\right] + \exp\left(\frac{2\lambda_{\mathrm{IG}}t}{\mu_{\mathrm{IG}}}\right)\Phi\left[-\sqrt{\frac{\lambda_{\mathrm{IG}}}{D}}\left(\frac{D}{\mu_{\mathrm{IG}}} + t\right)\right] \tag{6}$$

Considering the random effects, the parameters $\boldsymbol{\theta}_{\mathrm{IG}}$ are assumed to follow the Gamma distribution with hyper-parameters $\theta_{\mathrm{IG}}^{\mathrm{H}} = \left(\delta_{\mu_{\mathrm{IG}}}, \gamma_{\mu_{\mathrm{IG}}}, \delta_{\lambda_{\mathrm{IG}}}, \gamma_{\lambda_{\mathrm{IG}}}\right)$, as

$$\begin{cases} \mu_{\mathrm{IG}} \sim \mathrm{Ga}\left(\delta_{\mu_{\mathrm{IG}}}, \gamma_{\mu_{\mathrm{IG}}}\right) \\ \lambda_{\mathrm{IG}} \sim \mathrm{Ga}(\delta_{\lambda_{\mathrm{IG}}}, \gamma_{\lambda_{\mathrm{IG}}}) \end{cases} \tag{7}$$

### *2.2. Gamma process with random effects*

The PDF of Gamma distribution $y \sim \mathrm{Ga}(a_{\mathrm{Ga}}, b_{\mathrm{Ga}})$, $a_{\mathrm{Ga}}, b_{\mathrm{Ga}} > 0$, with mean and variance $a_{\mathrm{Ga}}b_{\mathrm{Ga}}$ and $a_{\mathrm{Ga}}b_{\mathrm{Ga}}^2$ respectively, is

$$f_{\mathrm{Ga}}(y|a_{\mathrm{Ga}}, b_{\mathrm{Ga}}) = \frac{1}{\Gamma(a_{\mathrm{Ga}}){b_{\mathrm{Ga}}}^{a_{\mathrm{Ga}}}} y^{a_{\mathrm{Ga}}-1} exp(-y/b_{\mathrm{Ga}}) \tag{8}$$

where $\Gamma(a) = \int_0^{+\infty} x^{a-1}\exp(-x)dx$ is the Gamma function.

Its CDF is

$$F_{\mathrm{Ga}}(y|a_{\mathrm{Ga}}, b_{\mathrm{Ga}}) = \frac{\varsigma(a_{\mathrm{Ga}}, y/b_{\mathrm{Ga}})}{\Gamma(a_{\mathrm{Ga}})} \tag{9}$$

where $\varsigma(a, b) = \int_0^b x^{a-1}\exp(-x)ds$ is the lower incomplete Gamma function.

A degradation process which can be described as Gamma process, has the following properties.

(a) The initial degradation $Y(0) = 0$ with probability 100%.
(b) $Y(t)$ has independent increments, $Y(t_4)$ - $Y(t_3)$ and $Y(t_2)$ - $Y(t_1)$ only if $t_4 > t_3 > t_2 > t_1$.
(c) The increments $\Delta Y(t) = Y(t+\Delta t) - Y(t)$ follow the Gamma distribution, as

$$\Delta Y(t) \sim \mathrm{Ga}(\mu_{\mathrm{Ga}}\Delta\Lambda(t), \lambda_{\mathrm{Ga}}) \tag{10}$$

with parameters $\theta_{\mathrm{Ga}} = (\mu_{\mathrm{Ga}}, \lambda_{\mathrm{Ga}})$, where $\mu_{\mathrm{Ga}} > 0$, $\lambda_{\mathrm{Ga}} > 0$, and where $\Delta\Lambda(t) = \Delta\Lambda(t+\Delta t) - \Lambda(t)$ and $\Lambda(t)$ is a monotonic increasing function.

The linear degradation processes are considered in this work, which allows the increments to be written as follows

$$\Delta Y(t) \sim \mathrm{Ga}(\mu_{\mathrm{Ga}} t, \lambda_{\mathrm{Ga}}) \tag{11}$$

Hence, the product degradation path $Y(t)$ has the following PDF,

$$f_{\mathrm{Ga}}(y(t)|\mu_{\mathrm{Ga}} t, \lambda_{\mathrm{Ga}}) = \frac{y^{\mu_{\mathrm{Ga}} t-1}(t)}{\Gamma(\mu_{\mathrm{Ga}} t)\lambda_{\mathrm{Ga}}{}^{\mu_{\mathrm{Ga}} t}}\exp\left(-\frac{y(t)}{\lambda_{\mathrm{Ga}}}\right) \tag{12}$$

Provided a predefined failure threshold $D$, the reliability can be represented as

$$R_{\mathrm{Ga}}(t|\mu_{\mathrm{Ga}} t, \lambda_{\mathrm{Ga}}) = 1 - \frac{\Gamma(\mu_{\mathrm{Ga}} t, D/\lambda_{\mathrm{Ga}})}{\Gamma(\mu_{\mathrm{Ga}} t)} \tag{13}$$

where $\Gamma(a, b) = \int_b^{+\infty} x^{a-1}\exp(-x)dx$ is the upper incomplete Gamma function.

Considering the random effects, the parameters $\boldsymbol{\theta}_{\mathrm{Ga}}$ are assumed to follow the Gamma distribution with hyper-parameters $\theta_{\mathrm{Ga}}^{\mathrm{H}} = \left(\delta_{\mu_{\mathrm{Ga}}}, \gamma_{\mu_{\mathrm{Ga}}}, \delta_{\lambda\mathrm{Ga}}, \gamma_{\lambda\mathrm{Ga}}\right)$, as

$$\begin{cases} \mu_{\mathrm{Ga}} \sim \mathrm{Ga}\left(\delta_{\mu_{\mathrm{Ga}}}, \gamma_{\mu_{\mathrm{Ga}}}\right) \\ \lambda_{\mathrm{Ga}} \sim \mathrm{Ga}(\delta_{\lambda\mathrm{Ga}}, \gamma_{\lambda\mathrm{Ga}}) \end{cases} \tag{14}$$

## 3. Model combination method

In this section, the BMA method is applied to combine the IG process and Gamma process and analyze the model uncertainty. To consider the unit-to-unit variations and heterogeneities within product population, the model probabilities are assumed to follow the TN distribution. The random effects of both model parameters and model probability are considered.

Generally, based on Bayesian inference, the posterior probability $P(M_c|\mathbf{Y})$ that the candidate model $M_c$ is the best approximation model for dataset $\mathbf{Y}$, can be obtained as follows

$$P(M_c|\mathbf{Y}) = \frac{P(M_c)L(M_c|\mathbf{Y})}{\sum_{i=1}^{C} P(M_i)L(M_i|\mathbf{Y})} \tag{15}$$

where $c$ is the number of candidate model, $L(M_c|\mathbf{Y})$ is the likelihood of model $M_c$, and $P(M_c)$ is the prior probability for model $M_c$, which is given as follows

$$P(M_c) = \begin{cases} p_{M_c} & \text{if} M_c \text{istrue} \\ 0 & \text{other} \end{cases} \quad 0 \le p_{M_c} \le 1, \sum_{c=1}^{C} p_{M_c} = 1 \tag{16}$$

where $p_{M_c}$ is the probability for model $M_c$, with $0 \le p_{M_c} \le 1$, $\sum_{c=1}^{C} p_{M_c} = 1$.

There are two candidate models, and the Eq. (15) can be rewritten as follows,

$$P(M_1|\mathbf{Y}) = \frac{P(M_1)L(M_1|\mathbf{Y})}{\sum_{i=1}^{2} P(M_i)L(M_i|\mathbf{Y})} \tag{17}$$

Here $M_1$ indicates the IG process and $M_2$ indicates the Gamma process. The probability for $M_1$, $P(M_1)$, and the probability for $M_2$, $P(M_2)$, satisfy the following relationship $P(M_1) + P(M_2) = 1$. Similarly, the posterior probabilities $P(M_1|\mathbf{Y})$ and $P(M_2|\mathbf{Y})$ satisfy Eq. (18).

$$P(M_1|\mathbf{Y}) + P(M_2|\mathbf{Y}) = 1 \tag{18}$$

Since the only other candidate model, besides the IG process, is the Gamma process, the probability for $M_2$ can be determined by the probability for $M_1$ based on Eq. (18). The random effects are introduced in the model probabilities by assuming that the parameter $\theta_{\mathrm{M}} = (p_{\mathrm{M}_1})$ follows TN distribution with the hyper-parameters $\theta_{\mathrm{M}}^{\mathrm{H}} = (\mu_{\mathrm{M}}, \sigma_{\mathrm{M}})$, as

$$p_{\mathrm{M}_1} \sim TN(u_{\mathrm{M}}, \sigma_{\mathrm{M}}, 0, 1) \tag{19}$$

The PDF of $p_{\mathrm{M}_1}$ is

$$f(p_{\mathrm{M}_1}|\mu_{\mathrm{M}}, \sigma_{\mathrm{M}}, 0, 1) = \begin{cases} \dfrac{\phi\left(\frac{p_{\mathrm{M}_1}-\mu_{\mathrm{M}}}{\sigma_{\mathrm{M}}}\right)}{\Phi\left(\frac{1-\mu_{\mathrm{M}}}{\sigma_{\mathrm{M}}}\right) - \Phi\left(\frac{-\mu_{\mathrm{M}}}{\sigma_{\mathrm{M}}}\right)} & 0 \le p_{\mathrm{M}_1} \le 1 \\ 0 & \text{other} \end{cases} \tag{20}$$

where $\phi(\cdot)$ is the PDF of the standard Gaussian distribution.

## 4. Parameters estimation

In this section, the Bayesian inference is applied to estimate the distribution hyper-parameters. To estimate the distribution hyper-parameters, three steps need to be performed, as shown in Fig. 2, Where $\mathbf{Y}_{mn}$ is the degradation observation dataset, $\boldsymbol{Y}_i$ is the vector of

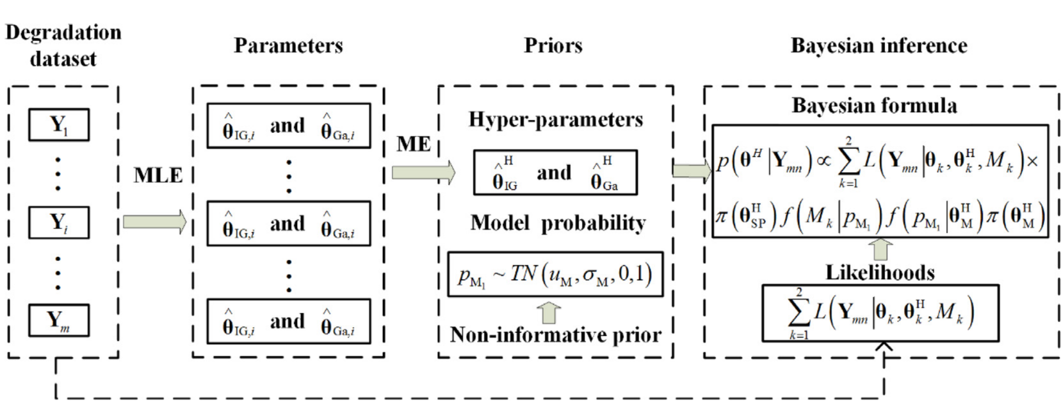


**Fig. 2.** Schematic representation of the parameters estimation method.

degradation observations for $i$-th tested sample, $\hat{\theta}_{\mathrm{IG,i}}$ and $\hat{\theta}_{\mathrm{Ga,i}}$ are vectors of parameters' MLE results for $i$th tasted sample, $\hat{\theta}_{\mathrm{IG}}^{\mathrm{H}}$ and $\hat{\theta}_{\mathrm{Ga}}^{\mathrm{H}}$ are the vectors of ME results of hyper-parameters, and $P_{M1}$ is the model probability. Firstly, the parameters' priors are obtained by ME combined with MLE for each candidate model. The MLE is used to estimate each candidate model parameters for each tested sample, and then the hyper-parameters for each candidate model are estimated by ME and based on the MLE results. Secondly, the non-informative prior is adopted for the model prior probabilities. Lastly, the hyper-parameters are obtained by fully Bayesian inference method.

### 4.1. Parameter priors

In the presented work, degradation is determined based on a uniform observation time interval and is represented by the degradation observations matrix $\mathbf{Y}_{mn}$ of the following form,

$$\mathbf{Y}_{mn} = \begin{bmatrix} y_{1,1} & y_{1,2} & \cdots & y_{1,n} \\ y_{2,1} & y_{2,2} & \cdots & y_{2,n} \\ \vdots & \vdots & \ddots & \vdots \\ y_{m,1} & y_{m,2} & \cdots & y_{m,n} \end{bmatrix} \tag{21}$$

where $m$ is the number of the tested samples, $n$ is the number of the measurements for each tested sample, and $y_{i,j}$ presents the degradation observation at $j$th measurement for $i$th tested sample.

The degradation observations vector $\mathbf{Y}_i$, for each tested sample, can be represented as

$$\mathbf{Y}_i = [y_{i,1}, y_{i,2}, \dots y_{i,n}] \tag{22}$$

The likelihood functions of IG process and Gamma process for $i$th tested sample can be represented as

$$L(\mathbf{Y}_i|\theta_{\mathrm{IG}}, M_{\mathrm{IG}}) = \prod_{j=1}^{n} \sqrt{\frac{\lambda_{\mathrm{IG}} t_j^2}{2\pi y_{i,j}^3}} \exp\left[-\frac{\lambda_{\mathrm{IG}}(y_{i,j} - \mu_{\mathrm{IG}} t_j)^2}{2\mu_{\mathrm{IG}}^2 y_{i,j}}\right] \tag{23}$$

$$L(\mathbf{Y}_i|\theta_{\mathrm{Ga}}, M_{\mathrm{Ga}}) = \prod_{j=1}^{n} \frac{y_{i,j}^{\mu_{\mathrm{Ga}} t_j - 1}}{\Gamma(\mu_{\mathrm{Ga}} t_j)\lambda_{\mathrm{Ga}}^{\mu_{\mathrm{Ga}} t_j}} \exp\left(-\frac{y_{i,j}}{\lambda_{\mathrm{Ga}}}\right) \tag{24}$$

Degradation of a tested sample is measured at uniform observation time interval $T$, so the $j$th observation time can be presented as $t_j = jT$. The likelihood functions for IG process and Gamma process can then be rewritten and represented in the form of Eqs. (25) and (26), respectively.

$$L(\mathbf{Y}_i|\theta_{\mathrm{IG}}, M_{\mathrm{IG}}) = \prod_{j=1}^{n} \sqrt{\frac{\lambda_{\mathrm{IG}} (jT)^2}{2\pi y_{i,j}^3}} \exp\left[-\frac{\lambda_{\mathrm{IG}}(y_{i,j} - j\mu_{\mathrm{IG}} T)^2}{2\mu_{\mathrm{IG}}^2 y_{i,j}}\right] \tag{25}$$

$$L(\mathbf{Y}_i|\theta_{\mathrm{Ga}}, M_{\mathrm{Ga}}) = \prod_{j=1}^{n} \frac{y_{i,j}^{j\mu_{\mathrm{Ga}} T - 1}}{\Gamma(j\mu_{\mathrm{Ga}} T)\lambda_{\mathrm{Ga}}^{j\mu_{\mathrm{Ga}} T}} \exp\left(-\frac{y_{i,j}}{\lambda_{\mathrm{Ga}}}\right) \tag{26}$$

The corresponding log-likelihood functions can be derived by

$$l(\theta_{\mathrm{IG}}|\mathbf{Y}_i, M_{\mathrm{IG}}) = \sum_{j=1}^{n} \left\{\frac{1}{2}\ln(\lambda_{\mathrm{IG}}) + \ln(jT) - \frac{1}{2}\ln(2\pi) - \frac{3}{2}\ln(y_{i,j}) - \frac{\lambda_{\mathrm{IG}}(y_{i,j} - j\mu_{\mathrm{IG}} T)^2}{2\mu_{\mathrm{IG}}^2 y_{i,j}}\right\} \tag{27}$$

$$l(\theta_{\mathrm{Ga}}|\mathbf{Y}_i, M_{\mathrm{Ga}}) = \sum_{j=1}^{n} \left\{(j\mu_{\mathrm{Ga}} T - 1)\ln(y_{i,j}) - \ln(\Gamma(j\mu_{\mathrm{Ga}} T)) - j\mu_{\mathrm{Ga}} T \ln(\lambda_{\mathrm{Ga}}) - \frac{y_{i,j}}{\lambda_{\mathrm{Ga}}}\right\} \tag{28}$$

The estimated system parameters $\hat{\theta}_{\mathrm{IG},i} = \left(\hat{\mu}_{\mathrm{IG},i}, \hat{\lambda}_{\mathrm{IG},i}\right)$ and $\hat{\theta}_{\mathrm{Ga},i} = \left(\hat{\mu}_{\mathrm{Ga},i}, \hat{\lambda}_{\mathrm{Ga},i}\right)$, represent MLE results of model parameters $\theta_{\mathrm{IG}} = (\mu_{\mathrm{IG}}, \lambda_{\mathrm{IG}})$ and $\theta_{\mathrm{Ga}} = (\mu_{\mathrm{Ga}}, \lambda_{\mathrm{Ga}})$ for the $i$th tested sample. They can be obtained by maximizing the likelihood functions of IG process, Eq. (27), and Gamma process, Eq. (28). The corresponding MLEs are shown in Appendix A. The priors of the model hyper-parameters are obtained from the mean values and variances of the above model parameters' MLE results, according to ME algorithm. In Section 2.1 the parameter $\mu_{\mathrm{IG}}$ is assumed to follow the $\mathrm{Ga}\left(\delta_{\mu_{\mathrm{IG}}}, \gamma_{\mu_{\mathrm{IG}}}\right)$, so its expectation and variance can be represented by Eqs. (29) and (30), respectively.

$$\begin{cases} E(\mu_{\mathrm{IG}}) = \delta_{\mu_{\mathrm{IG}}}\gamma_{\mu_{\mathrm{IG}}} \\ Var(\mu_{\mathrm{IG}}) = \delta_{\mu_{\mathrm{IG}}}\gamma_{\lambda_{\mathrm{IG}}}^2 \end{cases} \tag{29}$$

$$\begin{cases} E(\mu_{\mathrm{IG}}) = \frac{1}{m}\sum_{i=1}^{m}\hat{\mu}_{\mathrm{IG},i} \\ Var(\mu_{\mathrm{IG}}) = \frac{1}{m}\sum_{i=1}^{m}\left(\hat{\mu}_{\mathrm{IG},i} - E(\mu_{\mathrm{IG}})\right)^2 \end{cases} \tag{30}$$

The ME results for the hyper-parameters $\delta_{\mu_{\mathrm{IG}}}$ and $\gamma_{\mu_{\mathrm{IG}}}$ are given by

$$\begin{cases} \hat{\delta}_{\mu_{\mathrm{IG}}} = \dfrac{\left(\frac{1}{m}\sum_{i=1}^{m}\hat{\mu}_{\mathrm{IG},i}\right)^2}{\frac{1}{m}\sum_{i=1}^{m}\left(\hat{\mu}_{\mathrm{IG},i} - \frac{1}{m}\sum_{i=1}^{m}\hat{\mu}_{\mathrm{IG},i}\right)^2} \\ \hat{\gamma}_{\mu_{\mathrm{IG}}} = \dfrac{\frac{1}{m}\sum_{i=1}^{m}\left(\hat{\mu}_{\mathrm{IG},i} - \frac{1}{m}\sum_{i=1}^{m}\hat{\mu}_{\mathrm{IG},i}\right)^2}{\frac{1}{m}\sum_{i=1}^{m}\hat{\mu}_{\mathrm{IG},i}} \end{cases} \tag{31}$$

Similarly as the parameter $\mu_{\mathrm{IG}}$, the expectation and variance of parameter $\lambda_{\mathrm{IG}}$ can be represented by Eq. (32), and the ME results of the hyper-parameters $\delta_{\lambda_{\mathrm{IG}}}$ and $\gamma_{\lambda_{\mathrm{IG}}}$ are represented by Eq. (33).

$$\begin{cases} E(\lambda_{\mathrm{IG}}) = \delta_{\lambda_{\mathrm{IG}}}\gamma_{\lambda_{\mathrm{IG}}} \\ Var(\lambda_{\mathrm{IG}}) = \delta_{\mu_{\mathrm{IG}}}\gamma_{\lambda_{\mathrm{IG}}}^2 \\ E(\lambda_{\mathrm{IG}}) = \frac{1}{m}\sum_{i=1}^{m}\hat{\lambda}_{\mathrm{IG},i} \\ Var(\lambda_{\mathrm{IG}}) = \frac{1}{m}\sum_{i=1}^{m}\left(\hat{\lambda}_{\mathrm{IG},i} - E(\lambda_{\mathrm{IG}})\right)^2 \end{cases} \tag{32}$$

$$\begin{cases} \hat{\delta}_{\lambda_{\mathrm{IG}}} = \dfrac{\left(\frac{1}{m}\sum_{i=1}^{m}\hat{\lambda}_{\mathrm{IG},i}\right)^2}{\frac{1}{m}\sum_{i=1}^{m}\left(\hat{\lambda}_{\mathrm{IG},i} - \frac{1}{m}\sum_{i=1}^{m}\hat{\lambda}_{\mathrm{IG},i}\right)^2} \\ \hat{\gamma}_{\lambda_{\mathrm{IG}}} = \dfrac{\frac{1}{m}\sum_{i=1}^{m}\left(\hat{\lambda}_{\mathrm{IG},i} - \frac{1}{m}\sum_{i=1}^{m}\hat{\lambda}_{\mathrm{IG},i}\right)^2}{\frac{1}{m}\sum_{i=1}^{m}\hat{\lambda}_{\mathrm{IG},i}} \end{cases} \tag{33}$$

Similarly to the ME results of the hyper-parameters for IG process $\hat{\theta}_{\mathrm{IG}}^{\mathrm{H}} = \left(\hat{\delta}_{\mu_{\mathrm{IG}}}, \hat{\gamma}_{\mu_{\mathrm{IG}}}, \hat{\delta}_{\lambda_{\mathrm{IG}}}, \hat{\gamma}_{\lambda_{\mathrm{IG}}}\right)$, the ME results of the hyper-parameters for Gamma process $\hat{\theta}_{\mathrm{Ga}}^{\mathrm{H}} = \left(\hat{\delta}_{\mu_{\mathrm{Ga}}}, \hat{\gamma}_{\mu_{\mathrm{Ga}}}, \hat{\delta}_{\lambda_{\mathrm{Ga}}}, \hat{\gamma}_{\lambda_{\mathrm{Ga}}}\right)$ are represented as

$$\begin{cases} \hat{\delta}_{\mu_{\mathrm{Ga}}} = \dfrac{\left(\frac{1}{m}\sum_{i=1}^{m}\hat{\mu}_{\mathrm{Ga},i}\right)^2}{\frac{1}{m}\sum_{i=1}^{m}\left(\hat{\mu}_{\mathrm{Ga},i} - \frac{1}{m}\sum_{i=1}^{m}\hat{\mu}_{\mathrm{Ga},i}\right)^2}, & \hat{\gamma}_{\mu_{\mathrm{Ga}}} = \dfrac{\frac{1}{m}\sum_{i=1}^{m}\left(\hat{\mu}_{\mathrm{Ga},i} - \frac{1}{m}\sum_{i=1}^{m}\hat{\mu}_{\mathrm{Ga},i}\right)^2}{\frac{1}{m}\sum_{i=1}^{m}\hat{\mu}_{\mathrm{Ga},i}} \\ \hat{\delta}_{\lambda_{\mathrm{Ga}}} = \dfrac{\left(\frac{1}{m}\sum_{i=1}^{m}\hat{\lambda}_{\mathrm{Ga},i}\right)^2}{\frac{1}{m}\sum_{i=1}^{m}\left(\hat{\lambda}_{\mathrm{Ga},i} - \frac{1}{m}\sum_{i=1}^{m}\hat{\lambda}_{\mathrm{Ga},i}\right)^2}, & \hat{\gamma}_{\lambda_{\mathrm{Ga}}} = \dfrac{\frac{1}{m}\sum_{i=1}^{m}\left(\hat{\lambda}_{\mathrm{Ga},i} - \frac{1}{m}\sum_{i=1}^{m}\hat{\lambda}_{\mathrm{Ga},i}\right)^2}{\frac{1}{m}\sum_{i=1}^{m}\hat{\lambda}_{\mathrm{Ga},i}} \end{cases} \tag{34}$$

In this paper, the $s$-independent prior distributions are set based on the ME results of the hyper-parameters as Eqs. (35) and (36). The variances are set to be one-tenth of the ME results of the hyper-parameters for the illustrative example in Section 5; whereas in other cases, it can be set based on the standard error.

$$\begin{cases}\delta_{\mu_{\rm IG}} \sim {\rm TN}(\hat{\delta}_{\mu_{\rm IG}}, 0.01\hat{\delta}^2_{\mu_{\rm IG}}, 0, +\infty), \gamma_{\mu_{\rm IG}} \sim {\rm TN}\left(\hat{\gamma}_{\mu_{\rm IG}}, 0.01\hat{\gamma}^2_{\mu_{\rm IG}}, 0, +\infty\right)\\ \delta_{\lambda_{\rm IG}} \sim {\rm TN}(\hat{\delta}_{\lambda_{\rm IG}}, 0.01\hat{\delta}^2_{\lambda_{\rm IG}}, 0, +\infty), \gamma_{\lambda_{\rm IG}} \sim {\rm TN}(\hat{\gamma}_{\lambda_{\rm IG}}, 0.01\hat{\gamma}^2_{\lambda_{\rm IG}}, 0, +\infty)\end{cases} \tag{35}$$

$$\begin{cases}\delta_{\mu_{\rm Ga}} \sim {\rm TN}(\hat{\delta}_{\mu_{\rm Ga}}, 0.01\hat{\delta}^2_{\mu_{\rm Ga}}, 0, +\infty), \gamma_{\mu_{\rm Ga}} \sim {\rm TN}\left(\hat{\gamma}_{\mu_{\rm Ga}}, 0.01\hat{\gamma}^2_{\mu_{\rm Ga}}, 0, +\infty\right)\\ \delta_{\lambda_{\rm Ga}} \sim {\rm TN}(\hat{\delta}_{\lambda_{\rm Ga}}, 0.01\hat{\delta}^2_{\lambda_{\rm Ga}}, 0, +\infty), \gamma_{\lambda_{\rm Ga}} \sim {\rm TN}(\hat{\gamma}_{\lambda_{\rm Ga}}, 0.01\hat{\gamma}^2_{\lambda_{\rm Ga}}, 0, +\infty)\end{cases} \tag{36}$$

### 4.2. Model prior probability

The model prior probability and its hyper-parameters are generally determined based on previous degradation data and expert experience. However, in situations when the cases lack prior information, the non-informative prior is frequently used. Adopting the non-informative prior, the prior of the hyper-parameters $\theta_{\rm M}^{\rm H} = (\mu_{\rm M}, \sigma_{\rm M})$ can be set as shown in Eq. (37). This approach is used in example studied in Section 5.

$$\begin{cases}\mu_{\rm M} \sim {\rm TN}(0.5, 0.01, 0, 1)\\ \sigma_{\rm M} \sim {\rm Uniform}\ (0, 1)\end{cases} \tag{37}$$

### 4.3. Posterior

The fully Bayesian inference method is used to estimate the hyper-para-
meters-
$\theta^{\rm H} = (\theta_{\rm SP}^{\rm H}, \theta_{\rm M}^{\rm H}) = \left(\delta_{\mu_{\rm IG}}, \gamma_{\mu_{\rm IG}}, \delta_{\lambda_{\rm IG}}, \gamma_{\lambda_{\rm IG}}, \delta_{\mu_{\rm Ga}}, \gamma_{\mu_{\rm Ga}}, \delta_{\lambda_{\rm Ga}}, \gamma_{\lambda_{\rm Ga}}, \mu_{\rm M}, \sigma_{\rm M}\right)$, as shown in Eq. (38), where the system parameters are as follows: model probability parameter $\theta_{\rm M} = (p_{\rm M_1})$, the IG process model parameter $\theta_{\rm IG} = (\mu_{\rm IG}, \lambda_{\rm IG})$, the Gamma process model parameters$\theta_{\rm Ga} = (\mu_{\rm Ga}, \lambda_{\rm Ga})$, the degradation process model parameters $\theta_{\rm SP} = (\theta_{\rm IG}, \theta_{\rm Ga}) = (\mu_{\rm IG}, \lambda_{\rm IG}, \mu_{\rm Ga}, \lambda_{\rm Ga})$, and the model probability hyper-parameters $\theta_{\rm M}^{\rm H} = (\mu_{\rm M}, \sigma_{\rm M})$. The posterior distribution of the hyper-parameters $p(\theta^{\rm H}|\mathbf{Y}_{mn})$ can be obtained from Eq. (39), where the likelihood function is given by Eq. (40).

$$p(\theta^H|\mathbf{Y}_{mn}) = \frac{\sum_{k=1}^{2} L(Y|\theta_k, \theta_k^{\rm H}, M_k)\pi(\theta_{\rm SP}^{\rm H})f(M_k|p_{\rm M_1})f(p_{\rm M_1}|\theta_{\rm M}^{\rm H})\pi(\theta_{\rm M}^{\rm H})}{\int_{\theta_{\rm M}^{\rm H}}\int_{\theta_{\rm SP}^{\rm H}}\sum_{k=1}^{2} L(Y|\theta_k, \theta_k^{\rm H}, M_k)\pi(\theta_{\rm SP}^{\rm H})f(M_k|p_{\rm M_1})f(p_{\rm M_1}|\theta_{\rm M}^{\rm H})\pi(\theta_{\rm M}^{\rm H})d\theta_{\rm SP}^{\rm H}d\theta_{\rm M}^{\rm H}} \tag{38}$$

$$p(\theta^H|\mathbf{Y}_{mn}) \propto L\left(\mathbf{Y}_{mn}\middle|\delta_{\mu_{\rm IG}}, \gamma_{\mu_{\rm IG}}, \delta_{\lambda_{\rm IG}}, \gamma_{\lambda_{\rm IG}}, \delta_{\mu_{\rm Ga}}, \gamma_{\mu_{\rm Ga}}, \delta_{\lambda_{\rm Ga}}, \gamma_{\lambda_{\rm Ga}}, \mu_{\rm M}, \sigma_{\rm M}\right) \pi\left(\delta_{\mu_{\rm IG}}, \gamma_{\mu_{\rm IG}}, \delta_{\lambda_{\rm IG}}, \gamma_{\lambda_{\rm IG}}, \delta_{\mu_{\rm Ga}}, \gamma_{\mu_{\rm Ga}}, \delta_{\lambda_{\rm Ga}}, \gamma_{\lambda_{\rm Ga}}, \mu_{\rm M}, \sigma_{\rm M}\right) \tag{39}$$

$$L(\mathbf{Y}_{mn}|\theta^{\rm H}) = \prod_{j=1}^{n}\prod_{i=1}^{m}\left[\begin{array}{l}\sqrt{\frac{\lambda_{\rm IG}(jT)^2}{2\pi y_{i,j}^3}}\exp\left[-\frac{\lambda_{\rm IG}(y_{i,j}-\mu_{\rm IG}jT)^2}{2\mu_{\rm IG}^2 y_{i,j}}\right]p(M_1|p_{\rm M_1})\\ +\frac{y_{i,j}^{\mu_{\rm Ga}jT-1}}{\Gamma(\mu_{\rm Ga}jT)\lambda_{\rm Ga}{}^{\mu_{\rm Ga}jT}}\exp\left(-\frac{y_{i,j}}{\lambda_{\rm Ga}}\right)p(M_2|p_{\rm M_1})\end{array}\right]f(p_{\rm M_1}|\theta_{\rm M}^{\rm H}) f(\theta_{\rm SP}|\theta_{\rm SP}^{\rm H}) \tag{40}$$

The PDFs of degradation process model parameters $f(\theta_{\rm SP}|\theta_{\rm SP}^{\rm H})$ are presented in Appendix B. The posterior distributions of the hyper-parameters $p(\theta^{\rm H}|\mathbf{Y}_{mn})$ are obtained based on Markov Chain Monte Carlo (MCMC) simulation method using the software OpenBUGS. The reliability can be obtained from Eq. (41), and solved numerically using Monte Carlo method. The algorithm for Monte Carlo simulation is shown in Algorithm 1, where $N$ is the number of the samples in the Monte Carlo simulation, $T_{\rm up}$ is simulation length of degradation time, $\boldsymbol{\theta}^{\rm H}$ is the vector of the hyper-parameters, and $\Delta t$ is the simulation step size of degradation time.

$$R(t) = \int_{\theta_{\rm M}}\int_{\theta_{\rm SP}}\left[\begin{array}{l}\Phi\left[\sqrt{\frac{\lambda_{\rm IG}}{D}}\left(\frac{D}{\mu_{\rm IG}}-t\right)\right]p_{\rm M_1}\\ +\exp\left(\frac{2\lambda_{\rm IG}t}{\mu_{\rm IG}}\right)\Phi\left[-\sqrt{\frac{\lambda_{\rm IG}}{D}}\left(\frac{D}{\mu_{\rm IG}}+t\right)\right]p_{\rm M_1}\\ +\left(1-\frac{\Gamma\left(\mu_{\rm Ga}t, D/\lambda_{\rm Ga}\right)}{\Gamma\left(\mu_{\rm Ga}t\right)}\right)[1-p_{\rm M_1}]\end{array}\right]f(\theta_{\rm M}|\theta_{\rm M}^{\rm H})f(\theta_{\rm SP}|\theta_{\rm SP}^{\rm H})d\theta_{\rm M}d\theta_{\rm SP} \tag{41}$$

## 5. Model verification

A GaAs laser degradation dataset presented by Meeker and Escobar [50] is used to demonstrate distinguishing features of the proposed reliability analysis method for evaluation of population reliability. The GaAs laser degradation dataset, shown in Fig. 3, has been widely studied in reliability engineering, and it has been verified that both the IG process and Gamma process can be used to analyze the degradation of this particular degradation dataset [14,16]. Furthermore, the pre-defined failure threshold $D$ is set consistently with that set in Refs. [14] and [16]. The degradation tests were performed on 15 tested samples, where the operating current was selected as degradation indicator and was measured at 250 hours interval. The proposed BMA based model considering random effects of parameters and model probabilities is presented as REPMP model $M_{\rm REPMP}$. The BMA based model considering random effects of parameters is presented as REP model $M_{\rm REP}$. The BMA based model considering random effects of model probabilities is presented as REMP model $M_{\rm REMP}$. The BMA based model, which does not consider random effects, is presented as NRE model $M_{\rm NRE}$. The proposed REPMP model is compared with the REP, REMP, and NRE models.

To obtain the prior distribution of hyper-parameters, the model parameters for Gamma process and IG process are estimated based on MLE method, as described in Section 4.1, and the MLE results are shown in Table 1.

Based on Eqs. (31), (33), and (34), the ME results of the model hyper-parameters $\theta_{\rm SP}^{\rm H}$ are given by

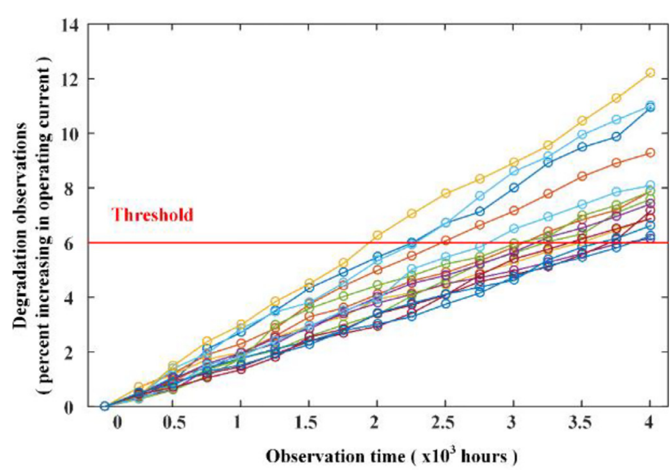


**Fig. 3.** The observations of GaAs laser degradation assuming the failure threshold is D = 6%.

**Table 1**
MLE results for Gamma process and IG process.

| Sample number | $\hat{\mu}_{\mathrm{IG},i}$ | $\hat{\lambda}_{\mathrm{IG},i}$ | $\hat{\mu}_{\mathrm{Ga},i}$ | $\hat{\lambda}_{\mathrm{Ga},i}$ | Sample number | $\hat{\mu}_{\mathrm{IG},i}$ | $\hat{\lambda}_{\mathrm{IG},i}$ | $\hat{\mu}_{\mathrm{Ga},i}$ | $\hat{\lambda}_{\mathrm{Ga},i}$ |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 2.6921 | 1.2622 | 0.9770 | 4.3509 | 9 | 1.9876 | 0.9343 | 0.8432 | 3.8389 |
| 2 | 2.4097 | 1.1330 | 0.9255 | 4.1412 | 10 | 3.0212 | 1.4171 | 1.0332 | 4.5800 |
| 3 | 1.8279 | 0.8547 | 0.8073 | 3.6942 | 11 | 1.9144 | 0.9000 | 0.8164 | 3.8300 |
| 4 | 1.7062 | 0.7965 | 0.7837 | 3.5996 | 12 | 2.0365 | 0.9520 | 0.8497 | 3.8760 |
| 5 | 1.8047 | 0.8434 | 0.8044 | 3.6712 | 13 | 2.0674 | 0.9671 | 0.8523 | 3.9085 |
| 6 | 2.7426 | 1.2851 | 0.9865 | 4.3867 | 14 | 1.6897 | 0.7913 | 0.7823 | 3.5727 |
| 7 | 1.6112 | 0.7559 | 0.7657 | 3.4999 | 15 | 1.6303 | 0.7659 | 0.7676 | 3.5174 |
| 8 | 1.5782 | 0.7399 | 0.7603 | 3.4650 | | | | | |
| Mean | 2.048 | 0.9599 | 0.8503 | 3.8621 | Variance | 0.1957 | 0.0432 | 0.0073 | 0.1158 |

$$\begin{cases} \hat{\delta}_{\mu_{\mathrm{IG}}} = 21.4323, \quad \hat{\gamma}_{\mu_{\mathrm{IG}}} = 0.0956 \\ \hat{\delta}_{\lambda_{\mathrm{IG}}} = 21.3289, \quad \hat{\gamma}_{\lambda_{\mathrm{IG}}} = 0.0450 \\ \hat{\delta}_{\mu_{\mathrm{Ga}}} = 99.0425, \quad \hat{\gamma}_{\mu_{\mathrm{Ga}}} = 0.0086 \\ \hat{\delta}_{\lambda_{\mathrm{Ga}}} = 128.8067, \quad \hat{\gamma}_{\lambda_{\mathrm{Ga}}} = 0.0300 \end{cases} \tag{42}$$

Based on the ME results of the model hyper-parameters $\theta_{\mathrm{SP}}^{\mathrm{H}}$, as described in Eqs. (35) and (36), the prior distributions of model hyper-parameters can be given by

$$\begin{cases} \delta_{\mu_{\mathrm{IG}}} \sim \mathrm{TN}(21.4323,\ 4.5934,\ 0,\ +\infty),\ \gamma_{\mu_{\mathrm{IG}}} \\ \quad \sim \mathrm{TN}(0.0956,\ 9.1394\times10^{-5},\ 0,\ +\infty) \\ \delta_{\lambda_{\mathrm{IG}}} \sim \mathrm{TN}(21.3289,\ 4.5492,\ 0,\ +\infty),\ \gamma_{\lambda_{\mathrm{IG}}} \\ \quad \sim \mathrm{TN}(0.0450,\ 2.025\times10^{-5},\ 0,\ +\infty) \\ \delta_{\mu_{\mathrm{Ga}}} \sim \mathrm{TN}(99.0425,\ 98.0948,\ 0,\ +\infty),\ \gamma_{\mu_{\mathrm{Ga}}} \\ \quad \sim \mathrm{TN}(0.0086,\ 7.396\times10^{-7},\ 0,\ +\infty) \\ \delta_{\lambda_{\mathrm{Ga}}} \sim \mathrm{TN}(128.8067,\ 165.9117,\ 0,\ +\infty),\ \gamma_{\lambda_{\mathrm{Ga}}} \\ \quad \sim \mathrm{TN}(0.0300,\ 9\times10^{-6},\ 0,\ +\infty) \end{cases} \tag{43}$$

For the proposed REPMP model, the posterior distributions of the hyper-parameters are obtained based on MCMC simulation method using the software OpenBUGS. The simulation results are shown in Table 2, and the mean values are chosen as the estimates of hyper-parameters. The calculated PDFs of the model parameters and model probability are shown in Fig. 4. It can be observed that the model probability parameter $p_1$ and model parameters are randomly distributed, and the variances and mean values are about the same size. Consequently, it can be concluded that it is necessary to consider the random effects in the model probability parameter and model parameters related to the unit-to-unit variations and heterogeneities within product population. Furthermore, all of the PDFs displayed in Fig. 2 are almost non-skew, it can be seen that, based on the proposed method, the median, mode and mean of the estimated values for the parameters are almost as the same values.

**Table 2**
MCMC simulation results of the hyper-parameters for the proposed REPMP model.

| Parameter | Mean (Estimation) | Sd | Median | 95% *s*-credibility interval |
|---|---|---|---|---|
| $\delta_{\mu_{\mathrm{IG}}}$ | 21.42 | 2.146 | 21.42 | [17.23,25.65] |
| $\gamma_{\mu_{\mathrm{IG}}}$ | 0.09558 | 0.009538 | 0.09555 | [0.07704,0.1144] |
| $\delta_{\lambda_{\mathrm{IG}}}$ | 21.33 | 2.141 | 21.33 | [17.14,25.54] |
| $\gamma_{\lambda_{\mathrm{IG}}}$ | 0.045 | 0.004507 | 0.04499 | [0.03617,0.05384] |
| $\delta_{\mu_{\mathrm{Ga}}}$ | 156.7 | 8.494 | 156.7 | [140.2,173.6] |
| $\gamma_{\mu_{\mathrm{Ga}}}$ | 0.01368 | $7.306\times10^{-4}$ | 0.01367 | [0.01226,0.01512] |
| $\delta_{\lambda_{\mathrm{Ga}}}$ | 59.92 | 18.42 | 56.21 | [33.99,102.8] |
| $\gamma_{\lambda_{\mathrm{Ga}}}$ | 0.01755 | 0.00488 | 0.01745 | [0.009082,0.02708] |
| $\mu_{\mathrm{M}}$ | 0.3098 | 0.08251 | 0.3095 | [0.148,0.472] |
| $\sigma_{\mathrm{M}}$ | 0.09143 | 0.06454 | 0.08111 | [0.004289,0.241] |

The simulation results of the hyper-parameters and parameters for REP, REMP, and NRE models are shown in Tables 3–5, respectively. The corresponding likelihood functions and reliability functions for REP, REMP, and NRE models are given in Appendix C.

The corresponding log-likelihood functions of the *i*th tested sample for the proposed REPMP, REP, REMP, and NRE models are given in Appendix D. The minus log-likelihood is used to reflect the likelihood of the model, and the smaller value of minus log-likelihood implies the greater possibility. The calculation results of the minus log-likelihood for each test sample and model are shown in Fig. 5. It can be observed that the proposed REPMP model results in the minimum value of minus log-likelihood for each tested sample. Namely, the REPMP is the best approximation model for the analyzed degradation dataset.

To compare these models, the minus log-likelihood values and the variances of the minus log-likelihood values are calculated, and the results are presented in Table 6 for each model. It can be observed that the minus log-likelihood value of the proposed REPMP model is the lowest, indicating that the goodness-of-fit of the proposed REPMP model is the best one among these models. Furthermore, the variance of the minus log-likelihood values for the proposed REPMP model is also the lowest, which indicates that the proposed REPMP model is more flexible in evaluating the population reliability compared to other models.

The minus log-likelihood value can only be used to reflect the goodness-of-fit but cannot necessarily reflect the complexity of the model, which cannot be ignored in model selection. Generally, the AIC and BIC are frequently used to describe the appropriateness of the model for fitting a dataset, considering both the goodness-of-fit and complexity of the model. Furthermore, the AIC provides a criterion to estimate both the goodness of fit and estimation complexity of the selected model, while the BIC not only considers these two factors but also includes the effects of the sample number. Hence, the AIC and BIC are also introduced to discuss the effectiveness of different models. The definition of AIC value is $\mathrm{AIC} = 2k - 2l(\hat{\theta})$, whereas definition of BIC value is $\mathrm{BIC} = k\ln(mn) - 2l(\hat{\theta})$, where $k$ is the number of parameters in the model, $\hat{\theta}$ represents estimated values of the parameters, and $l(\hat{\theta})$ is the value of the log-likelihood. The calculated results for AIC and BIC values are presented in Table 6. Whether the AIC or BIC is used in model selection, the proposed REPMP model results in the best fit for the analyzed GaAs laser degradation dataset. Based on the previously discussed results, it is apparent that the proposed REPMP model has a number of advantages compared to other models, including the best goodness-of-fit not only for each tested sample but also for the degradation dataset. The proposed model also provides the flexibility for the population reliability evaluation, and it still performs well

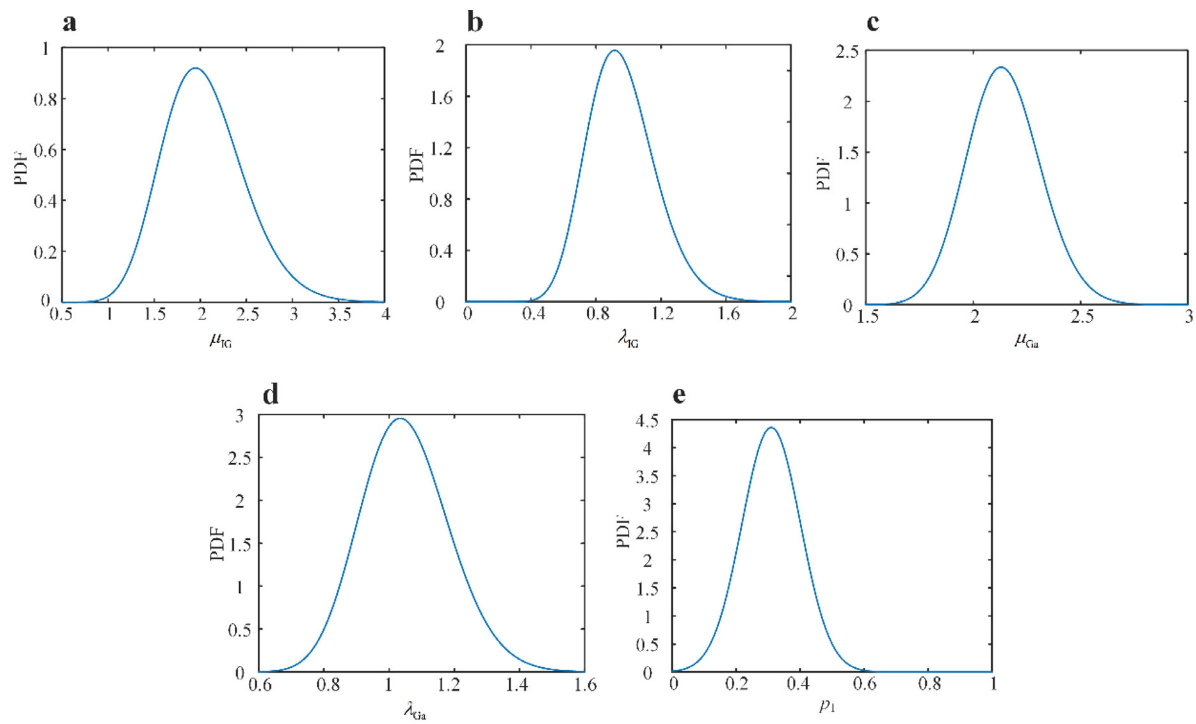


**Fig. 4.** The PDFs of the model parameters and model probability.

**Table 3**
MCMC simulation results for the REP model.

| Parameter | Mean | Sd | Median | 95% *s*-credibility interval | Prior distribution |
|---|---|---|---|---|---|
| $\delta_{\mu IG}$ | 21.42 | 2.139 | 21.43 | (17.23,25.63) | TN(21.4324,4.5934,0,+∞) |
| $\gamma_{\mu IG}$ | 0.09555 | 0.009523 | 0.09559 | (0.07679,0.1142) | TN(0.0956,$9.1394 \times 10^{-5}$,0,+∞) |
| $\delta_{\lambda IG}$ | 21.3 | 2.139 | 21.29 | (17.1,25.49) | TN(21.3289,4.5492,0,+∞) |
| $\gamma_{\lambda IG}$ | 0.04493 | 0.004504 | 0.04493 | (0.03607,0.05375) | TN(0.0450,$2.025 \times 10^{-5}$,0,+∞) |
| $\delta_{\mu Ga}$ | 156.9 | 8.516 | 156.9 | (140.3,173.7) | TN(99.0425,98.0948,0,+∞) |
| $\gamma_{\mu Ga}$ | 0.01369 | $7.365 \times 10^{-4}$ | 0.01368 | (0.01226,0.01515) | TN(0.0086,$7.396 \times 10^{-7}$,0,+∞) |
| $\delta_{\lambda,Ga}$ | 59.89 | 18.46 | 56.07 | (33.95,103.1) | TN(128.8067,165.9117,0,+∞) |
| $\gamma_{\lambda Ga}$ | 0.01754 | 0.004882 | 0.01746 | (0.008999,0.02707) | TN(0.03,$9 \times 10^{-6}$,0,+∞) |
| $p_1$ | 0.3332 | 0.2357 | 0.2926 | (0.0125,0.8421) | Uniform(0,1) |

**Table 4**
MCMC simulation results for the REMP model.

| Parameter | Mean | Sd | Median | 95% *s*-credibility interval | Prior distribution |
|---|---|---|---|---|---|
| $\mu_{IG}$ | 1.907 | 0.09847 | 1.904 | (1.725,2.109) | TN(2.048,0.1957,0,+∞) |
| $\lambda_{IG}$ | 2.539 | 0.1706 | 2.539 | (2.207,2.876) | TN(0.9599,0.0432,0,+∞) |
| $\mu_{Ga}$ | 0.9345 | 0.075 | 0.9348 | (0.7861,1.08) | TN(0.8503,0.0073,0,+∞) |
| $\lambda_{Ga}$ | 3.518 | 0.3443 | 3.508 | (2.871,4.22) | TN(3.8621,0.1158,0,+∞) |
| $\mu_M$ | 0.6175 | 0.08028 | 0.6177 | (0.4599,0.7742) | TN(0.5,0.01,0,+∞) |
| $\sigma_M$ | 0.1146 | 0.07751 | 0.1045 | (0.005371,0.2889) | Uniform(0,1) |

**Table 5**
MCMC simulation results for the NRE model.

| Parameter | Mean | Sd | Median | 95% *s*-credibility interval | Prior distribution |
|---|---|---|---|---|---|
| $\mu_{IG}$ | 2.054 | 0.446 | 2.047 | (1.194,2.96) | TN(2.048,0.1957,0,+∞) |
| $\lambda_{IG}$ | 0.9562 | 0.2078 | 0.9585 | (0.5495,1.362) | TN(0.9599,0.0432,0,+∞) |
| $\mu_{Ga}$ | 1.365 | 0.06742 | 1.365 | (1.237,1.501) | TN(0.8503,0.0073,0,+∞) |
| $\lambda_{Ga}$ | 1.547 | 0.0971 | 1.542 | (1.367,1.747) | TN(3.8621,0.1158,0,+∞) |
| $p_1$ | 0.3334 | 0.2356 | 0.2926 | (0.01284,0.8417) | Uniform(0,1) |

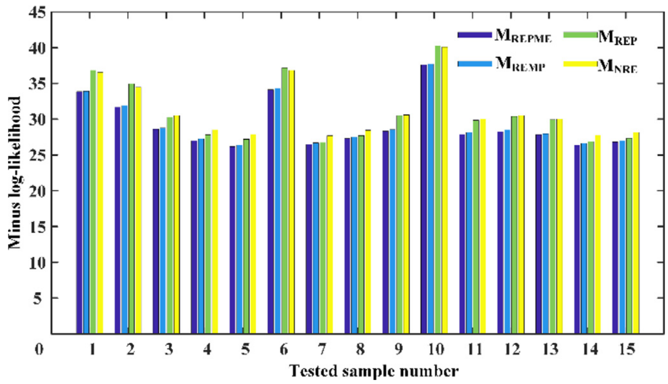


**Fig. 5.** Comparisons of minus log-likelihood values of each tested sample under different models.

**Table 6**
Goodness-of-fit comparisons between four models.

| | $M_{REPMP}$ | $M_{REP}$ | $M_{REMP}$ | $M_{NRE}$ |
|---|---|---|---|---|
| Minus log-likelihood | **438.5291** | 441.3068 | 463.9562 | 468.3134 |
| Variance of minus log-likelihood | **3.5777** | 3.5960 | 4.0207 | 3.8223 |
| AIC | **897.0582** | 900.6136 | 939.9124 | 946.6268 |
| BIC | **931.8582** | 931.9336 | 960.7924 | 964.0268 |

considering both the goodness-of-fit and complexity. Therefore, it can be concluded that for evaluating the population reliability, the random effects need to be introduced into both the model probabilities and model parameters in order to consider the unit-to-unit variations and heterogeneities within product population.

**Algorithm 1**
Monte Carlo simulation for the reliability of the proposed model.

**Set** $N$, $T_{up}$, $\boldsymbol{\theta}^{H}$, $\Delta t$
**While** $t < T_{up}$
**Increase** $t$ with $\Delta t$
**For** $i = 1{:}N$
Sample a $\boldsymbol{\theta}_i$ from the PDFs for model parameters (shown in Appendix B) based on the $\boldsymbol{\theta}^{H}$.
Sample a $p_{M_{1i}}$ model probability (Eq. (20)) based on the $\boldsymbol{\theta}^{H}$.
**End For**
**For** $i = 1{:}N$
**Calculate** $R_{IGi}(t)$ by
$$R_{IG\,i}(t)=\Phi\left[\sqrt{\frac{\lambda_{IG\,i}}{D}}\left(\frac{D}{\mu_{IG\,i}}-t\right)\right]+\exp\left(\frac{2\lambda_{IG\,i}t}{\mu_{IG\,i}}\right)\Phi\left[-\sqrt{\frac{\lambda_{IG\,i}}{D}}\left(\frac{D}{\mu_{IG\,i}}+t\right)\right]$$
**Calculate** $R_{Gai}(t)$ by
$$R_{Ga\,i}(t)=1-\frac{\Gamma(\mu_{Ga\,i}t, D/\lambda_{Ga\,i})}{\Gamma(\mu_{Ga\,i}t)}$$
**Calculate** $R_i(t)$ by
$$R_i(t)=p_{M_1\,i}R_{IG\,i}(t)+[1-p_{M_1\,i}]R_{IG\,i}(t)$$
**End For**
**Calculate** $R(t)$ by
$$R(t)=\frac{1}{N}\sum_{i=1}^{N}R_i(t)$$
**Sort** $R_1(t),\ldots R_i(t),\ldots R_N(t)$
**Calculate** the 95% $s$-credibility intervals of reliability $R(t)$
**End While**

The predictions of the degradation path calculated by different models are shown in Fig. 6, the 95% $s$-credibility intervals and the mean values for the proposed REPMP model, REP model, REMP model and NRE model are calculated based on Eq. (41), Eq. (C2), Eq. (C4) and Eq. (C6), separately. The 95% $s$-credibility interval ranges, calculated by different models, are shown in Fig. 7. It can be observed that the maximum 95% $s$-credibility interval range is for the proposed REPMP model while it is the minimum for the NRE model. It can been observed

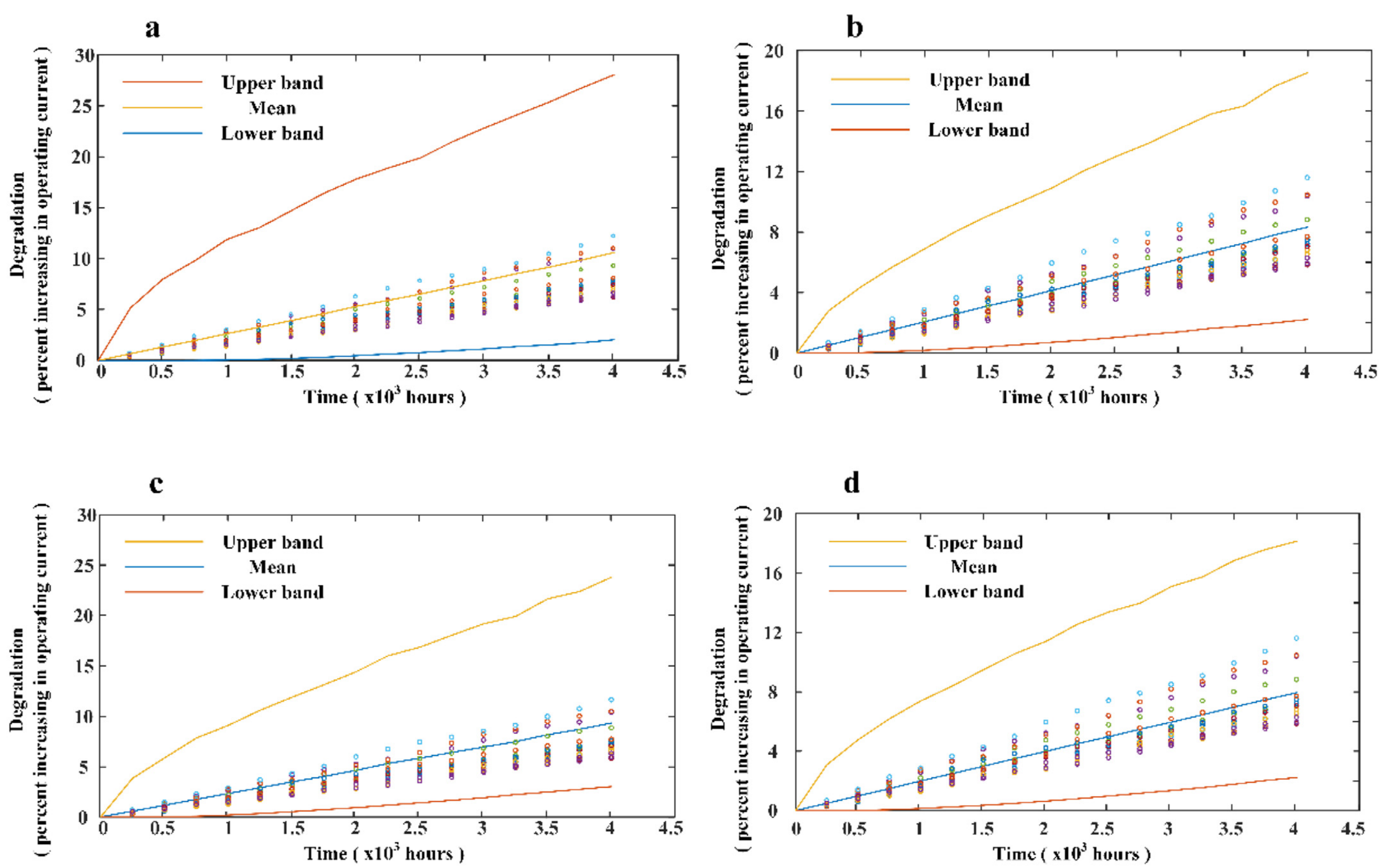


**Fig. 6.** Comparisons between the predictions calculated by different models, (a) the proposed REPMP model, (b) REP model, (c) REMP model and (d) NRE model.

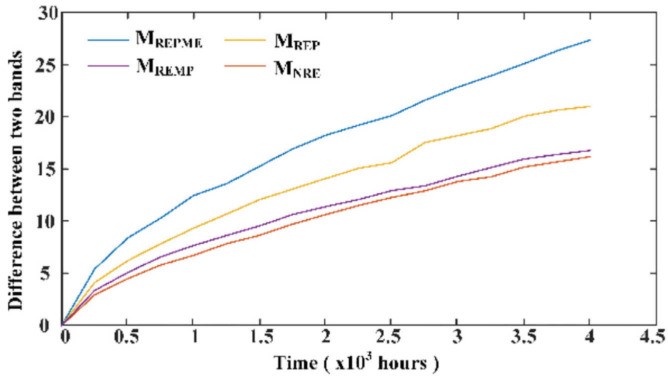


**Fig. 7.** Comparison of the 95% s-credibility ranges calculated by different models.

that for the population degradation predictions, the over-confident inference problem cannot be avoided regardless whether the random effects of model parameters or random effects of model probability are included or neglected.

Fig. 8 shows the comparisons between the PDF variations of reliability calculated by REPMP model, REP model, and REMP model. It can be observed that the reliabilities are randomly distributed at each observation time due to the random effects of model parameters, model probabilities, or both.

The 95% *s*-credibility ranges and the mean values of the predicted reliability for the REPMP, REP, and REMP models are shown in Fig. 9. The 95% *s*-credibility ranges calculated by different models are shown in Fig. 10. It can been observed that similarly to the degradation predictions, the over-confident inference problem cannot be avoided in the population reliability evaluation, regardless at whether the random effects of model parameters or random effects of model probability are included or neglected.

Furthermore, as shown in Table 6 and Figs. 7–10, compared with REMP model and NRE model, the Goodness-of-fit and predictions of REP model are closer to the REPMP model. It should be noted that the REMP model and NRE model neglect the random effects of model probabilities, while the NRE model and REP model neglect the random effects of model parameters. So it can be observed that in this case the random effects of model parameters impact on the predictions more significantly compared to the random effects of model probabilities.

## 6. Conclusions

A BMA based reliability analysis method for monotonic degradation dataset based on IG process and Gamma process with random effects is proposed. The Gamma process and IG process are selected as the candidate models, and the BMA method is used to combine these processes to handle the model uncertainty. The random effects of the model parameters and model possibility are taken into account in order to consider the unit-to-unit variations and heterogeneities within product population. Bayesian inference is used to estimate distribution hyperparameters in which the priors are obtained by ME combined with MLE.

A GaAs laser degradation dataset is used to demonstrate the effectiveness of the proposed REPMP method. The proposed BMA based reliability analysis method has a number of promising aspects compared to other models discussed in this paper. Specifically, the proposed REPMP model has the best goodness-of-fit not only for each tested sample but also for the degradation dataset. Comparing the AIC values and BIC values, the proposed REPMP model still provides the best results even when the complexity of the model is considered. The random effects in model probabilities and model parameters are also considered, and the results indicate that the proposed REPMP model is flexible enough to evaluate the population reliability.

Further work needs to address other aspects of applying stochastic processes in evaluation of reliability. For example, for model uncertainty, both the model averaging method and the adjustment factor

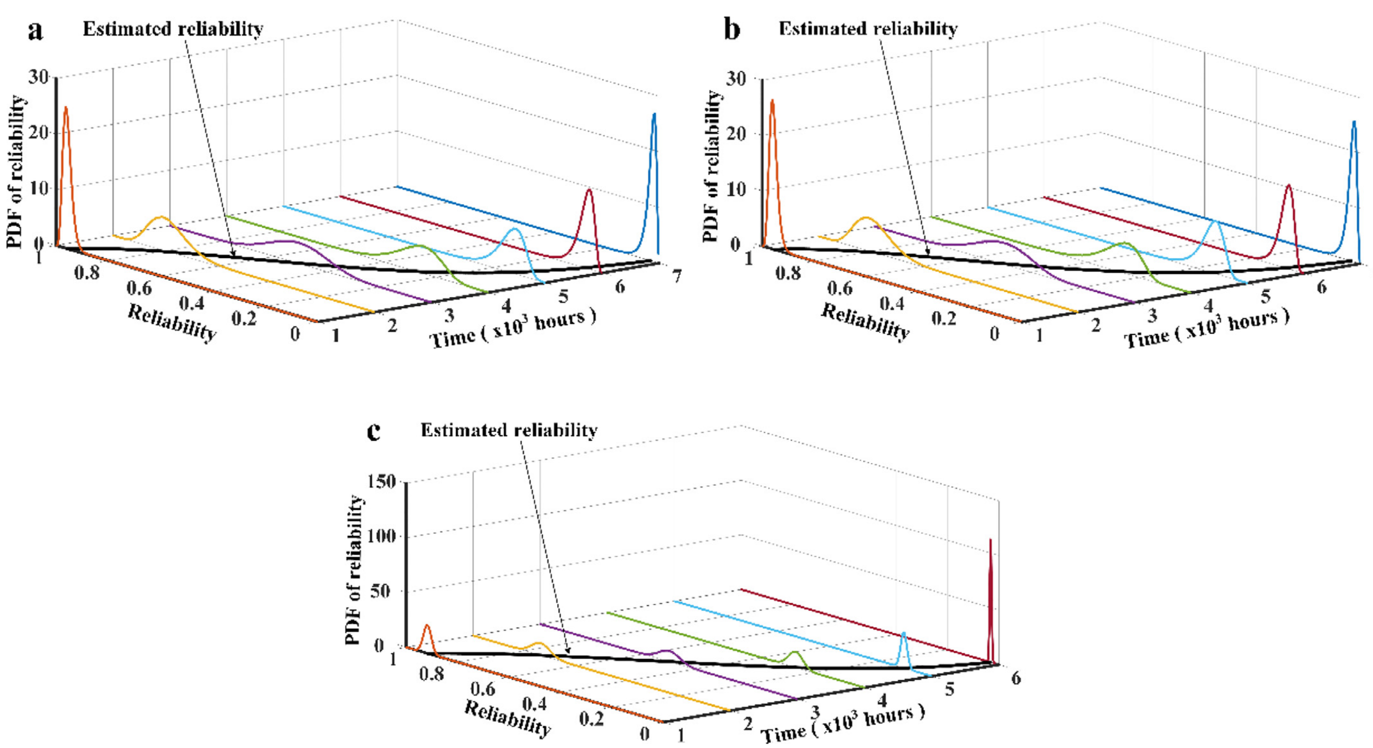


**Fig. 8.** Comparisons between the PDF of reliability calculated by (a) REPMP model, (b) REP model, and (c) REMP model.

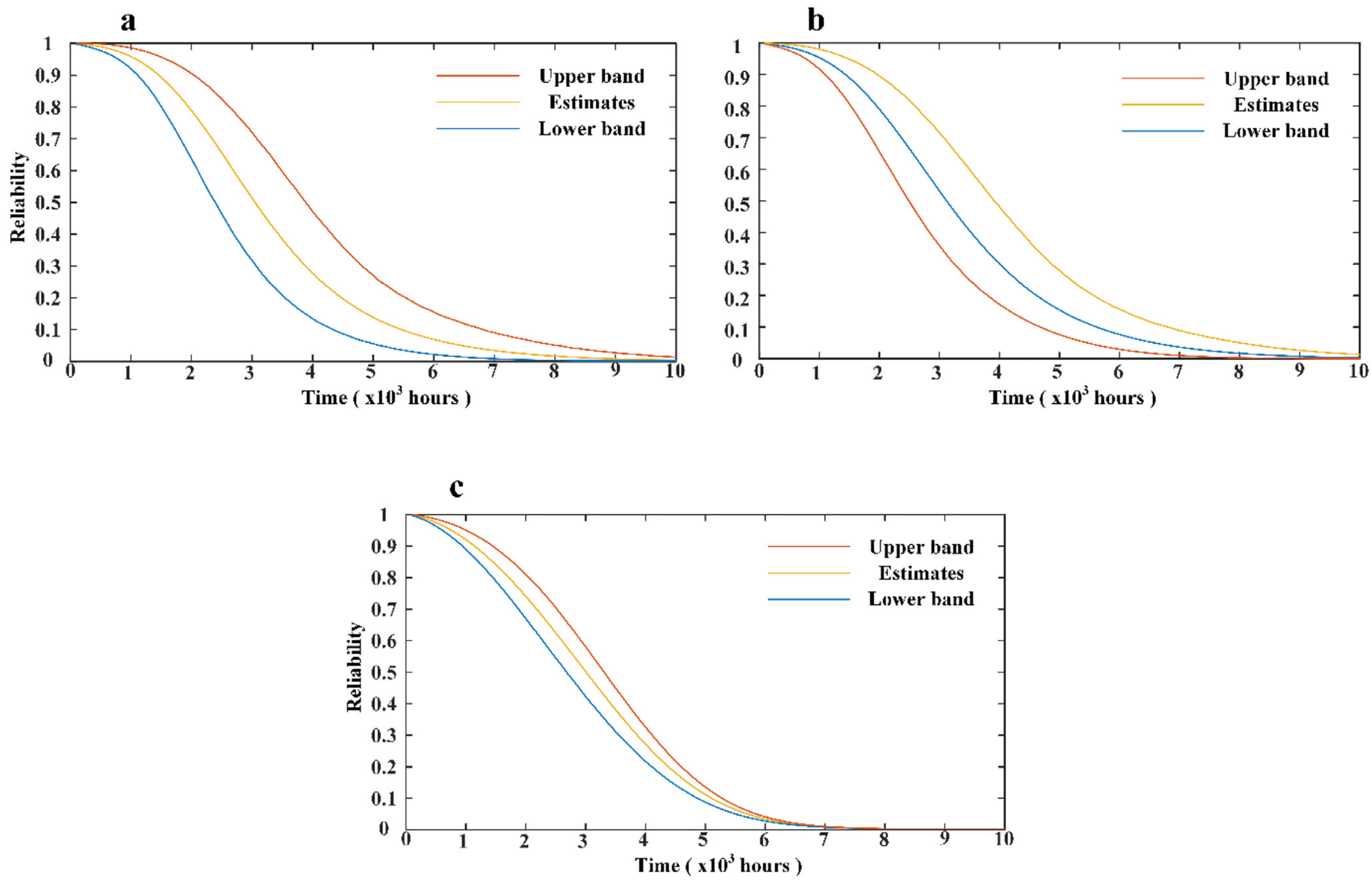


**Fig. 9.** Comparisons between the 95% *s*-credibility ranges of reliability calculated by (a) REPMP model, (b) REP model, and (c) REMP model.

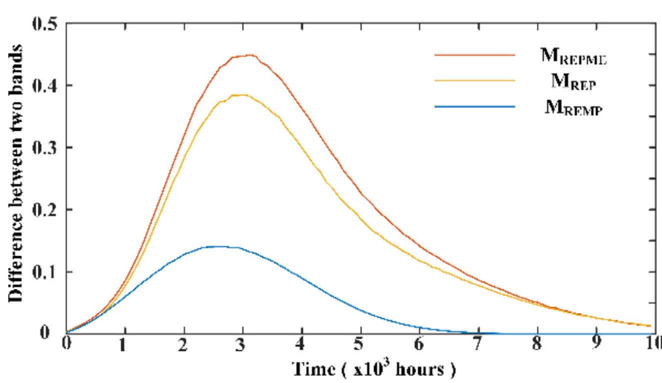


**Fig. 10.** Comparison of the widths of the 95% *s*-credibility ranges of reliability calculated by different models.

method need to assume that the degradation process is restricted in the model set with 100% probability. The problem of relaxing this hypothesis is very difficult. Although the D-S theory does not require this hypothesis, it is still difficult to elicit the belief values based on the expert knowledge to evaluate the model possibility.

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## Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.ress.2018.06.019.

## Appendix A. The MLE of the model parameters.

The MLE of IG process

The derivatives of corresponding log-likelihood function Eq. (27) to the model parameters $\theta_{IG} = (\mu_{IG}, \lambda_{IG})$ are given by

$$\begin{cases} \frac{\partial l(\theta_{IG}|Y_i, M_{IG})}{\partial \lambda_{IG}} = \frac{A_{IG}}{\lambda_{IG}} - \frac{B_{IG}}{\mu_{IG}^2} + \frac{C_{IG}}{\mu_{IG}} - D_{IG} \\ \frac{\partial l(\theta_{IG}|Y_i, M_{IG})}{\partial \mu_{IG}} = \frac{B_{IG}\lambda_{IG}}{\mu_{IG}^3} - \frac{C_{IG}\lambda_{IG}}{\mu_{IG}^2} \end{cases} \tag{A.1}$$

where

$$\begin{cases} A_{IG} = \frac{n}{2} \\ B_{IG} = \sum_{j=1}^{n} y_{i,j} \\ C_{IG} = T\sum_{j=1}^{n} j \\ D_{IG} = \frac{T^2}{2}\sum_{j=1}^{n}\frac{j^2}{y_{i,j}} \end{cases} \tag{A.2}$$

The MLE of model parameters $\theta_{IG} = (\mu_{IG}, \lambda_{IG})$ are

$$\begin{cases} \hat{\mu}_{IG,i} = \frac{B_{IG}}{C_{IG}} \\ \hat{\lambda}_{IG,i} = \frac{A_{IG}}{D_{IG}} \end{cases} \tag{A.3}$$

The MLE of Gamma process

Derivatives of the corresponding log-likelihood function Eq. (28) to the model parameters $\theta_{Ga} = (\mu_{Ga}, \lambda_{Ga})$ are given by

$$\begin{cases} \frac{\partial l(\theta_{Ga}|Y_i, M_{Ga})}{\partial \mu_{Ga}} = A_{Ga} - B_{Ga}\sum_{j=1}^{n} j\psi(j\mu_{Ga}T) - C_{Ga}\ln(\lambda_{Ga}) \\ \frac{\partial l(\theta_{Ga}|Y_i, M_{Ga})}{\partial \lambda_{Ga}} = -\frac{C_{Ga}\mu_{Ga}}{\lambda_{Ga}} + \frac{2D_{Ga}}{\lambda_{Ga}^2} \end{cases} \tag{A.4}$$

where

$$\begin{cases} A_{Ga} = T\sum_{j=1}^{n} j\ln(y_{i,j}) \\ B_{Ga} = T \\ C_{Ga} = T\sum_{j=1}^{n} j \\ D_{Ga} = \sum_{j=1}^{n} y_{i,j} \end{cases} \tag{A.5}$$

where $\psi(x) = \frac{\Gamma'(x)}{\Gamma(x)}$ and $\Gamma'(x)$ is the derivative of Gamma function.

The MLE results of the Gamma process parameters $\theta_{Ga} = (\mu_{Ga}, \lambda_{Ga})$ are obtained by numerically solving the following equations.

$$\begin{cases} 0 = A_{Ga} - B_{Ga}\sum_{j=1}^{n} j\psi(j\mu_{Ga}T) - C_{Ga}\ln(\lambda_{Ga}) \\ 0 = -\frac{C_{Ga}\mu_{Ga}}{\lambda_{Ga}} + \frac{2D_{Ga}}{\lambda_{Ga}^2} \end{cases} \tag{A.6}$$

## Appendix B. The PDFs for model parameters

Since the *s*-independent prior distributions are used in this work, the PDF of the model parameters vector can be given as

$$f(\theta_{SP}|\theta_{SP}^{H}) = f\left(\mu_{IG}\middle|\delta_{\mu_{IG}}, \gamma_{\mu_{IG}}\right) f(\lambda_{IG}|\delta_{\lambda_{IG}}, \gamma_{\lambda_{IG}}) f\left(\mu_{Ga}\middle|\delta_{\mu_{Ga}}, \gamma_{\mu_{Ga}}\right) f(\lambda_{Ga}|\delta_{\lambda_{Ga}}, \gamma_{\lambda_{Ga}}) \tag{B.1}$$

where

$$f\left(\mu_{IG}\middle|\delta_{\mu_{IG}}, \gamma_{\mu_{IG}}\right) = \begin{cases} \frac{1}{\Gamma(\delta_{\mu_{IG}})\gamma_{\mu_{IG}}{}^{\delta_{\mu_{IG}}}} y^{\delta_{\mu_{IG}}-1} exp\left(-\mu_{IG}/\gamma_{\mu_{IG}}\right) & 0 \le \mu_{IG} \\ 0 & \text{other} \end{cases} \tag{B.2}$$

$$f(\lambda_{IG}|\delta_{\lambda_{IG}}, \gamma_{\lambda_{IG}}) = \begin{cases} \frac{1}{\Gamma(\delta_{\lambda_{IG}})\gamma_{\lambda_{IG}}{}^{\delta_{\lambda_{IG}}}} y^{\delta_{\lambda_{IG}}-1} exp(-\lambda_{IG}/\gamma_{\lambda_{IG}}) & 0 \le \lambda_{IG} \\ 0 & \text{other} \end{cases} \tag{B.3}$$

$$f\left(\mu_{Ga}\middle|\delta_{\mu_{Ga}}, \gamma_{\mu_{Ga}}\right) = \begin{cases} \frac{1}{\Gamma(\delta_{\mu_{Ga}})\gamma_{\mu_{Ga}}{}^{\delta_{\mu_{Ga}}}} y^{\delta_{\mu_{Ga}}-1} exp\left(-\mu_{Ga}/\gamma_{\mu_{Ga}}\right) & 0 \le \mu_{Ga} \\ 0 & \text{other} \end{cases} \tag{B.4}$$

$$f(\lambda_{Ga}|\delta_{\lambda_{Ga}}, \gamma_{\lambda_{Ga}}) = \begin{cases} \frac{1}{\Gamma(\delta_{\lambda_{Ga}})\gamma_{\lambda_{Ga}}{}^{\delta_{\lambda_{Ga}}}} y^{\delta_{\lambda_{Ga}}-1} exp(-\lambda_{Ga}/\gamma_{\lambda_{Ga}}) & 0 \le \lambda_{Ga} \\ 0 & \text{other} \end{cases} \tag{B.5}$$

## Appendix C. The likelihood functions and reliability functions for the REP, REMP, and NRE models

The likelihood and reliability functions for the REP model are given by

$$L(\mathbf{Y}_{mn}|\theta_{\mathrm{SP}}^{\mathrm{H}},\theta_{\mathrm{M}}) = \prod_{j=1}^{n}\prod_{i=1}^{m}\left[\begin{array}{l}\sqrt{\frac{\lambda_{\mathrm{IG}}(jT)^2}{2\pi y_{i,j}^3}}\exp\left[-\frac{\lambda_{\mathrm{IG}}(y_{i,j}-\mu_{\mathrm{IG}}jT)^2}{2\mu_{\mathrm{IG}}^2 y_{i,j}}\right]p(M_1|p_{\mathrm{M}_1})\\ +\frac{y_{i,j}^{\mu_{\mathrm{Ga}}jT-1}}{\Gamma(\mu_{\mathrm{Ga}}jT)\lambda_{\mathrm{Ga}}{}^{\mu_{\mathrm{Ga}}jT}}\exp\left(-\frac{y_{i,j}}{\lambda_{\mathrm{Ga}}}\right)p(M_2|p_{\mathrm{M}_1})\end{array}\right]f(\theta_{\mathrm{SP}}|\theta_{\mathrm{SP}}^{\mathrm{H}}) \tag{C.1}$$

$$R(t) = \int_{\theta_{\mathrm{SP}}}\left[\begin{array}{l}\Phi\left[\sqrt{\frac{\lambda_{\mathrm{IG}}}{D}}\left(\frac{D}{\mu_{\mathrm{IG}}}-t\right)\right]p_{\mathrm{M}_1}+\exp\left(\frac{2\lambda_{\mathrm{IG}}t}{\mu_{\mathrm{IG}}}\right)\Phi\left[-\sqrt{\frac{\lambda_{\mathrm{IG}}}{D}}\left(\frac{D}{\mu_{\mathrm{IG}}}+t\right)\right]p_{\mathrm{M}_1}\\ +\left(1-\frac{\Gamma\left(\mu_{\mathrm{Ga}}t, D/\lambda_{\mathrm{Ga}}\right)}{\Gamma\left(\mu_{\mathrm{Ga}}t\right)}\right)[1-p_{\mathrm{M}_1}]\end{array}\right]f(\theta_{\mathrm{SP}}|\theta_{\mathrm{SP}}^{\mathrm{H}})d\theta_{\mathrm{SP}} \tag{C.2}$$

The likelihood and reliability functions for the REMP model are given by

$$L(\mathbf{Y}_{mn}|\theta_{\mathrm{M}}^{\mathrm{H}},\theta_{\mathrm{SP}}) = \prod_{j=1}^{n}\prod_{i=1}^{m}\left[\begin{array}{l}\sqrt{\frac{\lambda_{\mathrm{IG}}(jT)^2}{2\pi y_{i,j}^3}}\exp\left[-\frac{\lambda_{\mathrm{IG}}(y_{i,j}-\mu_{\mathrm{IG}}jT)^2}{2\mu_{\mathrm{IG}}^2 y_{i,j}}\right]p(M_1|p_{\mathrm{M}_1})\\ +\frac{y_{i,j}^{\mu_{\mathrm{Ga}}jT-1}}{\Gamma(\mu_{\mathrm{Ga}}jT)\lambda_{\mathrm{Ga}}{}^{\mu_{\mathrm{Ga}}jT}}\exp\left(-\frac{y_{i,j}}{\lambda_{\mathrm{Ga}}}\right)p(M_2|p_{\mathrm{M}_1})\end{array}\right]f(p_{\mathrm{M}_1}|\theta_{\mathrm{M}}^{\mathrm{H}}) \tag{C.3}$$

$$R(t) = \int_{\theta_{\mathrm{M}}^{\mathrm{H}}}\left[\begin{array}{l}\Phi\left[\sqrt{\frac{\lambda_{\mathrm{IG}}}{D}}\left(\frac{D}{\mu_{\mathrm{IG}}}-t\right)\right]p_{\mathrm{M}_1}+\exp\left(\frac{2\lambda_{\mathrm{IG}}t}{\mu}\right)\Phi\left[-\sqrt{\frac{\lambda_{\mathrm{IG}}}{D}}\left(\frac{D}{\mu_{\mathrm{IG}}}+t\right)\right]p_{\mathrm{M}_1}\\ +\left(1-\frac{\Gamma\left(\mu_{\mathrm{Ga}}t, D/\lambda_{\mathrm{Ga}}\right)}{\Gamma\left(\mu_{\mathrm{Ga}}t\right)}\right)[1-p_{\mathrm{M}_1}]\end{array}\right]f(\theta_{\mathrm{M}}|\theta_{\mathrm{M}}^{\mathrm{H}})d\theta_{\mathrm{M}}^{\mathrm{H}} \tag{C.4}$$

The likelihood and reliability functions for the NRE model are given by

$$L(\mathbf{Y}_{mn}|\theta) = \prod_{j=1}^{n}\prod_{i=1}^{m}\left[\begin{array}{l}\sqrt{\frac{\lambda_{\mathrm{IG}}(jT)^2}{2\pi y_{i,j}^3}}\exp\left[-\frac{\lambda_{\mathrm{IG}}(y_{i,j}-\mu_{\mathrm{IG}}jT)^2}{2\mu_{\mathrm{IG}}^2 y_{i,j}}\right]p(M_1|p_{\mathrm{M}_1})\\ +\frac{y_{i,j}^{\mu_{\mathrm{Ga}}jT-1}}{\Gamma(\mu_{\mathrm{Ga}}jT)\lambda_{\mathrm{Ga}}{}^{\mu_{\mathrm{Ga}}jT}}\exp\left(-\frac{y_{i,j}}{\lambda_{\mathrm{Ga}}}\right)p(M_2|p_{\mathrm{M}_1})\end{array}\right] \tag{C.5}$$

$$R(t) = \Phi\left[\sqrt{\frac{\lambda_{\mathrm{IG}}}{D}}\left(\frac{D}{\mu_{\mathrm{IG}}}-t\right)\right]p_{\mathrm{M}_1}+\exp\left(\frac{2\lambda_{\mathrm{IG}}t}{\mu_{\mathrm{IG}}}\right)\Phi\left[-\sqrt{\frac{\lambda_{\mathrm{IG}}}{D}}\left(\frac{D}{\mu_{\mathrm{IG}}}+t\right)\right]p_{\mathrm{M}_1} + \left(1-\frac{\Gamma\left(\mu_{\mathrm{Ga}}t, D/\lambda_{\mathrm{Ga}}\right)}{\Gamma\left(\mu_{\mathrm{Ga}}t\right)}\right)[1-p_{\mathrm{M}_1}] \tag{C.6}$$

## Appendix D. The log-likelihood functions of the *i*th tested sample

The log-likelihood functions of the *i*th tested sample for the proposed REPMP model is given by

$$l(\mathbf{Y}_i|\theta^{H}) = \sum_{j=1}^{n}\ln\left\{\left[\begin{array}{l}\sqrt{\frac{\lambda_{\mathrm{IG}}(jT)^2}{2\pi y_{i,j}^3}}\exp\left[-\frac{\lambda_{\mathrm{IG}}(y_{i,j}-\mu_{\mathrm{IG}}jT)^2}{2\mu_{\mathrm{IG}}^2 y_{i,j}}\right]p(M_1|p_{\mathrm{M}_1})\\ +\frac{y_{i,j}^{\mu_{\mathrm{Ga}}jT-1}}{\Gamma(\mu_{\mathrm{Ga}}jT)\lambda_{\mathrm{Ga}}{}^{\mu_{\mathrm{Ga}}jT}}\exp\left(-\frac{y_{i,j}}{\lambda_{\mathrm{Ga}}}\right)p(M_2|p_{\mathrm{M}_1})\end{array}\right]f(p_{\mathrm{M}_1}|\theta_{\mathrm{M}}^{\mathrm{H}})f(\theta_{\mathrm{SP}}|\theta_{\mathrm{SP}}^{\mathrm{H}})\right\} \tag{D.1}$$

The log-likelihood functions of the *i*th tested sample for REMP model is given by

$$l(\mathbf{Y}_i|\theta_{\mathrm{SP}}^{\mathrm{H}}) = \sum_{j=1}^{n}\ln\left\{\left[\begin{array}{l}\sqrt{\frac{\lambda_{\mathrm{IG}}(jT)^2}{2\pi y_{i,j}^3}}\exp\left[-\frac{\lambda_{\mathrm{IG}}(y_{i,j}-\mu_{\mathrm{IG}}jT)^2}{2\mu_{\mathrm{IG}}^2 y_{i,j}}\right]p(M_1|p_{\mathrm{M}_1})\\ +\frac{y_{i,j}^{\mu_{\mathrm{Ga}}jT-1}}{\Gamma(\mu_{\mathrm{Ga}}jT)\lambda_{\mathrm{Ga}}{}^{\mu_{\mathrm{Ga}}jT}}\exp\left(-\frac{y_{i,j}}{\lambda_{\mathrm{Ga}}}\right)p(M_2|p_{\mathrm{M}_1})\end{array}\right]f(\theta_{\mathrm{SP}}|\theta_{\mathrm{SP}}^{\mathrm{H}})\right\} \tag{D.2}$$

The log-likelihood functions of the *i*th tested sample for the REMP model is given by

$$l(\mathbf{Y}_i|\theta_{\mathrm{M}}^{\mathrm{H}}) = \sum_{j=1}^{n}\ln\left\{\left[\begin{array}{l}\sqrt{\frac{\lambda_{\mathrm{IG}}(jT)^2}{2\pi y_{i,j}^3}}\exp\left[-\frac{\lambda_{\mathrm{IG}}(y_{i,j}-\mu_{\mathrm{IG}}jT)^2}{2\mu_{\mathrm{IG}}^2 y_{i,j}}\right]p(M_1|p_{\mathrm{M}_1})\\ +\frac{y_{i,j}^{\mu_{\mathrm{Ga}}jT-1}}{\Gamma(\mu_{\mathrm{Ga}}jT)\lambda_{\mathrm{Ga}}{}^{\mu_{\mathrm{Ga}}jT}}\exp\left(-\frac{y_{i,j}}{\lambda_{\mathrm{Ga}}}\right)p(M_2|p_{\mathrm{M}_1})\end{array}\right]f(p_{\mathrm{M}_1}|\theta_{\mathrm{M}}^{\mathrm{H}})\right\} \tag{D.3}$$

The log-likelihood functions of the *i*th tested sample for the NRE model is given by

$$l(\mathbf{Y}_i|\theta) = \sum_{j=1}^{n} \ln\left[\begin{array}{l}\sqrt{\frac{\lambda_{\mathrm{IG}}(jT)^2}{2\pi y_{i,j}^3}}\exp\left[-\frac{\lambda_{\mathrm{IG}}(y_{i,j}-\mu_{\mathrm{IG}}jT)^2}{2\mu_{\mathrm{IG}}^2 y_{i,j}}\right]p(M_1|p_{\mathrm{M_1}}) \\ +\frac{y_{i,j}^{\mu_{\mathrm{Ga}}jT-1}}{\Gamma(\mu_{\mathrm{Ga}}jT)\lambda_{\mathrm{Ga}}{}^{\mu_{\mathrm{Ga}}jT}}\exp\left(-\frac{y_{i,j}}{\lambda_{\mathrm{Ga}}}\right)p(M_2|p_{\mathrm{M_1}})\end{array}\right] \tag{D.4}$$

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